

Dancing on the Saddles: A Geometric Framework for Stochastic Equilibrium Dynamics¹

Online Appendix

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A Sequential formulation of a canonical RBC

I consider a representative household with temporal log utility. Given initial condition (a_0, A_0) , the household maximizes lifetime utility under stochastic aggregate TFP A_t , subject to a budget constraint:

$$\max_{\{c_\tau(A^{(\tau)}), a_{\tau+1}(A^{(\tau)})\}_{\tau=0}^\infty} \mathbb{E}_0 \sum_{\tau=0}^{\infty} \beta^\tau \log(c_\tau(A^{(\tau)})) \quad (1)$$

$$\text{s.t. } c_\tau(A^{(\tau)}) + a_{\tau+1}(A^{(\tau)}) = a_\tau(A^{(\tau-1)})(1 + r(A^{(\tau)})) + w(A^{(\tau)}), \quad \forall \tau, \forall A^{(\tau)} \quad (2)$$

$$a_{\tau+1}(A^{(\tau)}) \geq -\bar{a}, \quad \forall \tau \quad (3)$$

where the superscript τ inside a bracket denotes the history of a variable up to period τ ; $-\bar{a}$ is the natural borrowing limit to preempt a Ponzi scheme. Labor supply is exogenously fixed at unity. I consider competitive factor prices given a CRS Cobb–Douglas production function:

$$r(A^{(\tau)}) = A_\tau \alpha (K(A^{(\tau)}))^{\alpha-1} - \delta \quad (4)$$

$$w(A^{(\tau)}) = A_\tau (1 - \alpha) (K(A^{(\tau)}))^\alpha, \quad (5)$$

K is the capital stock that satisfies $K(A^{(\tau)}) = a(A^{(\tau)})$ in equilibrium. With the regularity conditions given in [Stokey et al. \(1989\)](#), this sequential formulation yields the same optimality conditions as the recursive form in the main text.

B Individual conditional saddles in Aiyagari (1994)

I define individual-level conditional saddle paths in the heterogeneous-household economy without aggregate uncertainty ($A = A' = 1$). The conditional saddle is defined for the stationary RCE as in [Aiyagari \(1994\)](#).

Definition B.1 (Individual conditional saddle under frozen z).

Fix a stationary RCE of the Aiyagari (1994) economy, which delivers an individual asset policy

$$g_a : \mathbb{R}_+ \times \mathcal{Z} \rightarrow \mathbb{R}_+,$$

where $\mathcal{Z}$ is a finite Markov set with transition matrix Π . Fix an initial condition $(a_0, z_0) \in \mathbb{R}_+ \times \mathcal{Z}$. For any frozen idiosyncratic state $z \in \mathcal{Z}$, define the frozen- z continuation $\{a_t(z)\}_{t \geq 0}$

recursively by

$$a_{t+1}(z) = g_a(a_t(z), z), \quad a_0(z) = a_0. \quad (6)$$

The individual conditional saddle under frozen z is the orbit-closure

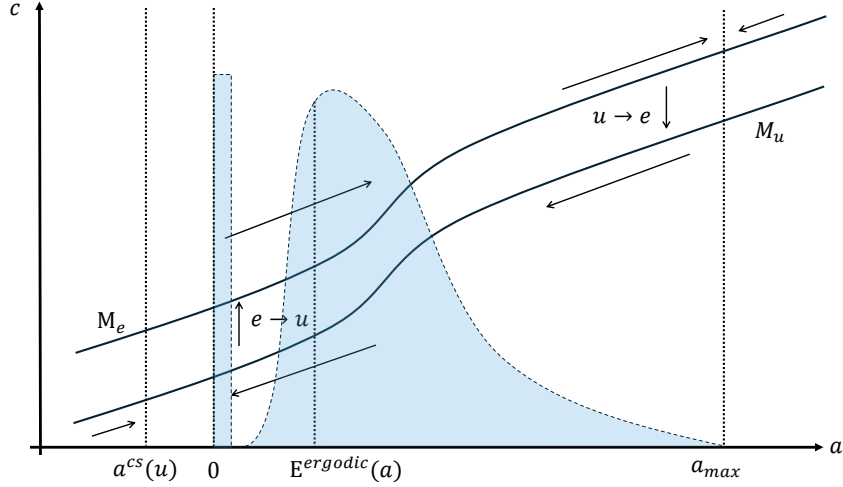
$$\mathcal{M}^{\text{ind}}(z; a_0) := \overline{\{a_t(z) : t \geq 0\}} \subseteq \mathbb{R}_+. \quad (7)$$

Definition B.2 (Individual conditional steady state under frozen z).

Fix (a_0, z_0) and $z \in \mathcal{Z}$, and let $\{a_t(z)\}_{t \geq 0}$ be the frozen- z continuation. The individual conditional steady state under frozen z is defined by

$$a^{cs}(z; a_0) := \lim_{t \rightarrow \infty} a_t(z). \quad (8)$$

Figure B.1: Individual conditional saddle paths in the stationary RCE



Notes: The figure illustrates the individual conditional saddle paths for $z = e$ and $z = u$ in Aiyagari (1994).

Figure B.1 illustrates individual-level conditional saddle paths in Aiyagari (1994). Because of the borrowing constraint, the conditional steady state associated with the unemployment state $z = u$ is not attained. Under standard calibrations, this generates a positive mass of agents at the borrowing limit. Analogous to the aggregate-level case, heterogeneity in the slopes of individual conditional saddle paths implies differential responses of individual consumption to idiosyncratic shocks.

C Proof of Proposition 4 (aggregate uncertainty and the conditional steady states)

Proposition C.1 (Aggregate uncertainty and the conditional steady states; restatement of Proposition 4).

Let $R(A, K) := 1 - \delta + \alpha AK^{\alpha-1}$ denote the gross return and $c(K, A)$ the stochastic-equilibrium consumption function. Suppose that (i) at each regime's PF steady state, the consumption response to a regime switch dominates the return response (jump dominance),

$$\frac{c(K_B^{pf}, G)}{c_B^{pf}} > \frac{R(G, K_B^{pf})}{R(B, K_B^{pf})}, \quad \frac{c(K_G^{pf}, B)}{c_G^{pf}} < \frac{R(B, K_G^{pf})}{R(G, K_G^{pf})},$$

and (ii) each conditional Euler residual F_A defined below is strictly positive for $K < K_A^{cs}$ and strictly negative for $K > K_A^{cs}$ on the relevant range (single crossing). Then

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}, \quad c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}.$$

Proof.

Let $c^{nul}(A, K) := AK^\alpha - \delta K$ denote the conditional K -nullcline. At any conditional or PF steady state, consumption lies on the nullcline: $c_A^{cs} = c(K_A^{cs}, A) = c^{nul}(A, K_A^{cs})$ and $c_A^{pf} = c^{nul}(A, K_A^{pf})$.

The PF steady states solve the Euler equations under absorbing beliefs,

$$1 = \beta R(B, K_B^{pf}), \quad 1 = \beta R(G, K_G^{pf}).$$

Since $R(A, K)$ is strictly increasing in A and strictly decreasing in K (because $\alpha - 1 < 0$), it follows immediately that $K_B^{pf} < K_G^{pf}$. This step is belief-free and uses neither hypothesis.

Next define the conditional Euler residuals at candidate K -stationary points, using the equilibrium post-switch consumption $c(K, A')$, exactly as in the conditional steady-state Euler equation of Section 2.2 of the main text:

$$F_B(K) := \beta \left[\pi_{BB} R(B, K) + \pi_{BG} \frac{c^{nul}(B, K)}{c(K, G)} R(G, K) \right] - 1,$$

$$F_G(K) := \beta \left[\pi_{GG} R(G, K) + \pi_{GB} \frac{c^{nul}(G, K)}{c(K, B)} R(B, K) \right] - 1.$$

By the conditional steady-state Euler equation, $F_B(K_B^{cs}) = 0$ and $F_G(K_G^{cs}) = 0$; hypothesis (ii) states that these are the unique zero crossings, from above.

Step 1: show $K_B^{cs} < K_B^{pf}$. Evaluate F_B at K_B^{pf} . Using $1 = \beta R(B, K_B^{pf})$, $\pi_{BB} = 1 - \pi_{BG}$, and $c^{nul}(B, K_B^{pf}) = c_B^{pf}$,

$$\begin{aligned} F_B(K_B^{pf}) &= \beta \left[(1 - \pi_{BG}) R(B, K_B^{pf}) + \pi_{BG} \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) \right] - 1 \\ &= \pi_{BG} \left[\beta \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) - 1 \right]. \end{aligned}$$

Jump dominance gives $c(K_B^{pf}, G) > c_B^{pf} R(G, K_B^{pf}) / R(B, K_B^{pf})$, hence

$$\beta \frac{c_B^{pf}}{c(K_B^{pf}, G)} R(G, K_B^{pf}) < \beta R(B, K_B^{pf}) = 1,$$

so the bracket is strictly negative and $F_B(K_B^{pf}) < 0$. By the single-crossing hypothesis, F_B is negative only above its zero, so $K_B^{cs} < K_B^{pf}$.

Step 2: show $K_G^{cs} > K_G^{pf}$. Symmetrically, using $1 = \beta R(G, K_G^{pf})$, $\pi_{GG} = 1 - \pi_{GB}$, and $c^{nul}(G, K_G^{pf}) = c_G^{pf}$,

$$F_G(K_G^{pf}) = \pi_{GB} \left[\beta \frac{c_G^{pf}}{c(K_G^{pf}, B)} R(B, K_G^{pf}) - 1 \right].$$

Jump dominance gives $c(K_G^{pf}, B) < c_G^{pf} R(B, K_G^{pf}) / R(G, K_G^{pf})$, hence

$$\beta \frac{c_G^{pf}}{c(K_G^{pf}, B)} R(B, K_G^{pf}) > \beta R(G, K_G^{pf}) = 1,$$

so $F_G(K_G^{pf}) > 0$. By the single-crossing hypothesis, F_G is positive only below its zero, so $K_G^{pf} < K_G^{cs}$.

Step 3: conclude the ordering and translate to consumption. Steps 1 and 2 have shown $K_B^{cs} < K_B^{pf}$ and $K_G^{pf} < K_G^{cs}$, and already $K_B^{pf} < K_G^{pf}$, hence

$$K_B^{cs} < K_B^{pf} < K_G^{pf} < K_G^{cs}.$$

Finally, along the K -nullcline $c^{nul}(A, K) = AK^\alpha - \delta K$ is strictly increasing in A and increasing in K on the dynamically efficient region ($\alpha AK^{\alpha-1} > \delta$, which holds at all four steady states under standard calibrations), so the same ordering carries over to consumption:

$$c_B^{cs} < c_B^{pf} < c_G^{pf} < c_G^{cs}.$$

■

Reading the hypotheses. Jump dominance compares two responses to the same regime switch at a PF steady state. The return ratio deviates from one only at second order in the regime gap, $R(G, K)/R(B, K) = 1 + (G - B)\alpha K^{\alpha-1}/R(B, K)$, whereas equilibrium consumption responds at first order to a persistent regime switch. Both hypotheses are directly checkable from the computed decision rules: condition (i) by evaluating $c(\cdot, A')$ at the two PF steady states, condition (ii) by evaluating F_A on a grid of K . In the baseline quarterly calibration ($\beta = 0.99$, $\alpha = 0.33$, $\delta = 0.025$, $A \in \{0.99, 1.01\}$, $\pi_{AA} = 0.90$), both hold with wide margins: at K_B^{pf} , consumption rises by 0.77% upon a switch to G against a return-ratio threshold of 0.07%, and at K_G^{pf} it falls by 0.76% against the same threshold; each residual F_A , evaluated with the computed decision rules, crosses zero exactly once on the computed saddle range, from above. The implied orderings are confirmed in the computed equilibrium, $K_B^{cs} = 27.53 < K_B^{pf} = 27.93 < K_G^{pf} = 28.77 < K_G^{cs} = 29.22$ (the PF-comparison figure in Section 2.2 of the main text).

Necessity of the strict conditions. The hypotheses cannot be dropped. Under full depreciation ($\delta = 1$) with log utility and Cobb–Douglas production, the equilibrium policy takes the Brock–Mirman form $K' = \alpha\beta AK^\alpha$ with $c = (1 - \alpha\beta)AK^\alpha$, independently of the transition matrix Π . Upon a regime switch at fixed K , consumption then moves exactly by the factor G/B , and so does the gross return $R(A, K) = \alpha AK^{\alpha-1}$: jump dominance holds with equality rather than strictly, and the conditional and PF steady states coincide, $K_A^{cs} = K_A^{pf} = (\alpha\beta A)^{1/(1-\alpha)}$. The strict orderings of Proposition 4 therefore genuinely require strict jump dominance; away from this degenerate corner, the conditions hold in the calibrated equilibrium and the strict inequalities obtain.

D Proof of Proposition 6 (state-dependent peak timing)

Proof.

The proof derives the peak time in closed form and applies the implicit function theorem to sign the comparative static in initial capital.

Step 1 (Blanchard–Kahn solution on the saddle branch). Log-linearize the RBC dynamics around the non-stochastic steady state. Let $\hat{k}_t := \ln K_t - \ln K^*$ and $\hat{a}_t := \ln A_t - \ln A^*$ denote

log-deviations. The linearized capital law of motion and saddle-path consumption policy are

$$\hat{k}_{t+1} = \lambda \hat{k}_t + \theta \hat{a}_t, \quad (9)$$

$$\hat{c}_t = \psi_k \hat{k}_t + \psi_a \hat{a}_t, \quad (10)$$

where $\lambda \in (0, 1)$ is the stable eigenvalue of the capital dynamics, $\theta > 0$ is the capital-TFP elasticity, and $\psi_k > 0$, $\psi_a > 0$ are the consumption-capital and consumption-TFP elasticities on the saddle branch.

Consider a one-time shock $\hat{a}_0 = \varepsilon > 0$ with deterministic mean-reversion $\hat{a}_t = \rho^t \varepsilon$, $\rho \in (0, 1)$. Iterating (9) gives

$$\hat{k}_t = \lambda^t \hat{k}_0 + \theta \varepsilon \sum_{s=0}^{t-1} \lambda^{t-1-s} \rho^s = \lambda^t \hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} (\lambda^t - \rho^t), \quad (11)$$

where the closed form uses $\lambda \neq \rho$. Substituting into (10):

$$\hat{c}_t = \psi_k \left[\lambda^t \hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} (\lambda^t - \rho^t) \right] + \psi_a \rho^t \varepsilon = \mathcal{P} \lambda^t + \mathcal{Q} \rho^t, \quad (12)$$

where

$$\mathcal{P}(\hat{k}_0) := \psi_k \left[\hat{k}_0 + \frac{\theta \varepsilon}{\lambda - \rho} \right], \quad \mathcal{Q} := \left[\psi_a - \frac{\psi_k \theta}{\lambda - \rho} \right] \varepsilon. \quad (13)$$

Crucially, $\mathcal{P}$ depends on $\hat{k}_0$ while $\mathcal{Q}$ does not.

Step 2 (Peak time in closed form). Treating t as continuous, the peak satisfies $d\hat{c}_t/dt = 0$:

$$\mathcal{P} \ln \lambda \cdot \lambda^{t^*} + \mathcal{Q} \ln \rho \cdot \rho^{t^*} = 0. \quad (14)$$

Rearranging:

$$\left(\frac{\lambda}{\rho} \right)^{t^*} = - \frac{\mathcal{Q} \ln \rho}{\mathcal{P} \ln \lambda}. \quad (15)$$

Taking logarithms yields

$$t^* = \frac{1}{\ln(\lambda/\rho)} \ln \left(\frac{-\mathcal{Q} \ln \rho}{\mathcal{P} \ln \lambda} \right), \quad (16)$$

which is well-defined when $-\mathcal{Q} \ln \rho / (\mathcal{P} \ln \lambda) > 0$, i.e., $\mathcal{Q}/\mathcal{P} < 0$ (since $\ln \lambda < 0$ and $\ln \rho < 0$ make the two negative signs cancel only when $\mathcal{Q}$ and $\mathcal{P}$ have opposite signs). With $\mathcal{P} > 0$,

this requires $\mathcal{Q} < 0$: the direct TFP effect on consumption is smaller than the capital-accumulation channel ($\psi_a < \psi_k \theta / (\lambda - \rho)$), which is exactly what generates the hump.

Step 3 (Comparative static via the implicit function theorem). Differentiating (16) with respect to $\hat{k}_0$:

$$\frac{\partial t^*}{\partial \hat{k}_0} = \frac{1}{\ln(\lambda/\rho)} \cdot \frac{-1}{\mathcal{P}} \cdot \frac{\partial \mathcal{P}}{\partial \hat{k}_0} = \frac{-\psi_k}{\mathcal{P}(\hat{k}_0) \cdot \ln(\lambda/\rho)}, \quad (17)$$

where $\partial \mathcal{P} / \partial \hat{k}_0 = \psi_k > 0$ from (13). Under $\lambda > \rho$, we have $\ln(\lambda/\rho) > 0$. Combined with $\psi_k > 0$ and $\mathcal{P} > 0$ (which holds when $\hat{k}_0 > -\theta\varepsilon/(\lambda - \rho)$, a condition easily satisfied within the ergodic capital range), this gives

$$\frac{\partial t^*}{\partial \hat{k}_0} < 0. \quad (18)$$

That is, the consumption peak arrives strictly later when initial capital is farther below the steady state. ■

E Proofs of Theorem 2 and Proposition 9 (synchronization and the across-spell bound)

Proof of Theorem 2 (synchronization under frozen-regime contraction).

Proof.

Part (i): existence of the pullback limit. Fix a regime history $\omega = (\dots, A_{t-1}, A_t)$ and, for $T \geq 0$, write the T -fold composition initialized at date $t - T$ as

$$G_T := \Gamma_{A_t} \circ \Gamma_{A_{t-1}} \circ \dots \circ \Gamma_{A_{t-T}} : \mathcal{U} \rightarrow \mathcal{U}, \quad (19)$$

which maps $\mathcal{U}$ into itself by forward invariance. For $T' > T$ and any $\Phi, \Phi' \in \mathcal{U}$,

$$d(G_T(\Phi), G_{T'}(\Phi')) = d(G_T(\Phi), G_T(\Psi)) \leq \lambda_c^{T+1} d(\Phi, \Psi) \leq \lambda_c^{T+1} \text{diam}(\mathcal{U}), \quad (20)$$

where $\Psi := (\Gamma_{A_{t-T-1}} \circ \dots \circ \Gamma_{A_{t-T'}})(\Phi') \in \mathcal{U}$ collects the extra maps of the longer composition; the contraction estimate applies because G_T is a composition of $T+1$ maps, each contracting at rate λ_c . Hence $\{G_T(\Phi)\}_{T \geq 0}$ is Cauchy uniformly in $\Phi \in \mathcal{U}$, and the limit

$$\Phi^*(\omega) := \lim_{T \rightarrow \infty} G_T(\Phi) \quad (21)$$

exists and is independent of Φ ; since $\mathcal{U}$ is closed in the complete space $(\mathcal{X}, d)$ and each $G_T(\Phi) \in \mathcal{U}$ by forward invariance, the limit lies in $\mathcal{U}$. Uniqueness follows from the same estimate: any state in $\mathcal{U}$ consistent with the full history ω is a pullback limit of itself and therefore equals $\Phi^*(\omega)$. This is the random-fixed-point construction of [Schenk-Hoppé and Schmalfuß \(2001\)](#) specialized to finitely many contracting maps selected by a Markov chain. Finally, write $H_T := \Gamma_{A_t} \circ \dots \circ \Gamma_{A_{t-T+1}}$ for the composition of the T maps applied strictly after the initialization date, so that a trajectory whose state at date $t - T$ is Φ arrives at $H_T(\Phi)$ at date t . By the cocycle identity $\Phi^*(\omega) = H_T(\Phi^*(\sigma^{-T}\omega))$, where $\sigma^{-T}\omega$ is the history shifted back T periods, and since H_T contracts at rate λ_c^T ,

$$d(H_T(\Phi), \Phi^*(\omega)) = d(H_T(\Phi), H_T(\Phi^*(\sigma^{-T}\omega))) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (22)$$

Part (ii): suffix synchronization. Suppose ω and ω' agree over the most recent T periods, and let $H_T := \Gamma_{A_t} \circ \dots \circ \Gamma_{A_{t-T+1}}$ denote the common composition of those T maps. Writing $\sigma^{-T}\omega$ for the history shifted back T periods, the pullback limits satisfy the cocycle identity $\Phi^*(\omega) = H_T(\Phi^*(\sigma^{-T}\omega))$ and likewise for ω' . Since both $\Phi^*(\sigma^{-T}\omega)$ and $\Phi^*(\sigma^{-T}\omega')$ lie in $\mathcal{U}$ (closedness, part (i)),

$$d(\Phi^*(\omega), \Phi^*(\omega')) = d(H_T(\Phi^*(\sigma^{-T}\omega)), H_T(\Phi^*(\sigma^{-T}\omega'))) \leq \lambda_c^T \text{diam}(\mathcal{U}). \quad (23)$$

■

Proof of Proposition 9 (across-spell approximation bound).

Proof.

Step 1 (Pathwise bound). By Theorem 2(i), each of the two states is the pullback limit of its history: $\Phi = \Phi^*(\omega)$, $\Phi' = \Phi^*(\omega')$. If the two histories agree over the most recent T periods, Theorem 2(ii) gives $d(\Phi, \Phi') \leq \lambda_c^T \text{diam}(\mathcal{U})$. This bound holds state by state, so it holds for every pair in the same- K cloud: fixing a common value of K selects a subset of history pairs and cannot break a pathwise inequality.

Step 2 (Distribution of the common suffix length). Let the two histories be drawn independently from the stationary regime chain, conditioned to agree at the current date (the current regime is commonly observed), so $T \geq 1$. The symmetric two-state chain is reversible, so its time reversal has the same transition matrix. Stepping backward one period, the two independent reversed chains continue to agree if both persist (probability ρ_A^2) or both switch

$((1 - \rho_A)^2)$, independently across periods. Hence, with

$$q := \rho_A^2 + (1 - \rho_A)^2 \in (1/2, 1), \quad (24)$$

the suffix length satisfies $\Pr(T = k) = q^{k-1}(1 - q)$ for $k = 1, 2, \dots$

Step 3 (Expected cloud diameter). Combining Steps 1 and 2,

$$\mathbb{E}[d(\Phi, \Phi')] \leq \text{diam}(\mathcal{U}) \mathbb{E}[\lambda_c^T] = \text{diam}(\mathcal{U}) \sum_{k=1}^{\infty} q^{k-1}(1 - q) \lambda_c^k = \frac{(1 - q) \lambda_c}{1 - q \lambda_c} \text{diam}(\mathcal{U}), \quad (25)$$

where the geometric series converges because $q \lambda_c < 1$. The factor is strictly below one and vanishes as $\rho_A \rightarrow 1$ (then $q \rightarrow 1$ and the expected suffix lengthens without bound) or as $\lambda_c \rightarrow 0$ (fast contraction within regimes). $\blacksquare$

F Proof of Proposition 10 (local contraction rate on the conditional saddle)

Regularity conditions. Fix $A \in \{B, G\}$ and assume the following at the conditional steady state Φ_A^{cs} :

- (i) (C^1 Banach structure) The distributional state space $\mathcal{X}$ admits a Banach chart in a neighborhood of Φ_A^{cs} , and the frozen-regime operator $\Gamma_{\text{endo}}(\cdot, A) : \mathcal{X} \rightarrow \mathcal{X}$ is continuously Fréchet-differentiable on that neighborhood; we write $D\Gamma_A := D\Gamma_{\text{endo}}(\Phi_A^{cs}, A)$. It suffices for the argument below that $\Gamma_{\text{endo}}(\cdot, A)$ be C^1 along $\mathcal{M}(\Phi_0, A)$, i.e., the composition with the parametrization of the conditional saddle in (iii) is C^1 .
- (ii) (Hyperbolicity along the saddle) The tangent to the conditional saddle $\mathcal{M}(\Phi_0, A)$ at Φ_A^{cs} is a one-dimensional invariant eigenspace of $D\Gamma_A$ with simple real eigenvalue $\lambda_A^s \in (0, 1)$, separated from the remainder of the spectrum by a spectral gap.
- (iii) (Invariant-curve regularity) In a neighborhood of Φ_A^{cs} there exists a C^1 one-dimensional locally invariant curve $\mathcal{W}_A \subset \mathcal{X}$, tangent at Φ_A^{cs} to that eigenspace, which contains the tail of the orbit: $\mathcal{M}(\Phi_0, A) \cap \mathcal{V} \subset \mathcal{W}_A$ for some neighborhood $\mathcal{V}$ of Φ_A^{cs} . (The orbit closure $\mathcal{M}(\Phi_0, A)$ itself is generically countable and is not required to be a manifold; the curve $\mathcal{W}_A$ is the strong-stable invariant manifold through Φ_A^{cs} .)
- (iv) (Metric compatibility) On the Banach chart of (i), the metric d is asymptotically

equivalent to the chart norm $\|\cdot\|$ near Φ_A^{cs} :

$$\lim_{\Phi, \Phi' \rightarrow \Phi_A^{cs}, \Phi \neq \Phi'} \frac{d(\Phi, \Phi')}{\|\Phi - \Phi'\|} = 1.$$

Full statement. Under (i)–(iv), for Φ, Φ' restricted to the conditional saddle $\mathcal{M}(\Phi_0, A)$,

$$\lim_{\substack{\Phi, \Phi' \rightarrow \Phi_A^{cs} \\ \Phi, \Phi' \in \mathcal{M}(\Phi_0, A), \Phi \neq \Phi'}} \frac{d(\Gamma_{\text{endo}}(\Phi, A), \Gamma_{\text{endo}}(\Phi', A))}{d(\Phi, \Phi')} = |\lambda_A^s|, \quad (26)$$

and for every $\varepsilon > 0$ there exists a neighborhood $\mathcal{U}_A^\varepsilon$ of Φ_A^{cs} such that $\Gamma_{\text{endo}}(\cdot, A)$ contracts on $\mathcal{M}(\Phi_0, A) \cap \mathcal{U}_A^\varepsilon$ at rate $|\lambda_A^s| + \varepsilon$.

Aggregate-capital projection. Because the aggregate map $K : \mathcal{X} \rightarrow \mathbb{R}$ is C^1 in the chart of (i) and the invariant curve $\mathcal{W}_A$ of assumption (iii) is tangent to the eigenspace spanned by e_A^s (with K nondegenerate along that direction), the curve $\mathcal{W}_A$ —and with it the orbit tail it contains—is locally parametrized by K near K_A^{cs} . The distributional contraction (26) therefore implies the standard log-linearization in capital units,

$$K_{t+1} - K_A^{cs} = \lambda_A^s (K_t - K_A^{cs}) + o(|K_t - K_A^{cs}|),$$

which becomes exact in the limit $K_t \rightarrow K_A^{cs}$. This is the form quoted in the main text.

Proof.

Work in the Banach chart of assumption (i), with norm $\|\cdot\|$. Let e_A^s denote a unit vector spanning the one-dimensional invariant eigenspace E_A^s of $D\Gamma_A$ (assumption (ii)), normalized so that $D\Gamma_A e_A^s = \lambda_A^s e_A^s$.

By assumption (iii), the invariant curve $\mathcal{W}_A$ is a C^1 one-dimensional curve tangent to E_A^s at Φ_A^{cs} that contains the orbit tail. Hence there exist a neighborhood of Φ_A^{cs} in the chart and a C^1 parametrization $\phi : (-\eta, \eta) \rightarrow \mathcal{W}_A$ with $\phi(0) = \Phi_A^{cs}$ and $\phi'(0) = e_A^s$; the orbit points near Φ_A^{cs} are of the form $\phi(s)$. For any two points $\Phi = \phi(s), \Phi' = \phi(s')$ on this local parametrization with $s \neq s'$,

$$\Phi - \Phi' = (s - s') e_A^s + r_1(s, s'), \quad \|r_1(s, s')\| = o(|s - s'|), \quad (27)$$

by the C^1 property of ϕ .

By assumption (i), $\Gamma_{\text{endo}}(\cdot, A)$ is C^1 in a neighborhood of Φ_A^{cs} (or, at minimum, C^1 along $\mathcal{M}(\Phi_0, A)$). Continuity of the derivative $D\Gamma_{\text{endo}}$ in Φ near Φ_A^{cs} upgrades the pointwise Fréchet

expansion at Φ_A^{cs} to a two-point expansion with uniform remainder: for Φ, Φ' in a small enough chart neighborhood,

$$\Gamma_{\text{endo}}(\Phi, A) - \Gamma_{\text{endo}}(\Phi', A) = D\Gamma_A(\Phi - \Phi') + r_2(\Phi, \Phi'), \quad \|r_2(\Phi, \Phi')\| = o(\|\Phi - \Phi'\|), \quad (28)$$

uniformly as $\Phi, \Phi' \rightarrow \Phi_A^{cs}$. (The single-point Fréchet expansion would control only $\Gamma(\Phi) - \Gamma(\Phi_A^{cs})$, not the two-point difference $\Gamma(\Phi) - \Gamma(\Phi')$ for arbitrary nearby pairs; this is why assumption (i) requires C^1 regularity on a neighborhood.) Substituting (27) and using $D\Gamma_A e_A^s = \lambda_A^s e_A^s$,

$$\Gamma_{\text{endo}}(\Phi, A) - \Gamma_{\text{endo}}(\Phi', A) = \lambda_A^s (s - s') e_A^s + o(|s - s'|). \quad (29)$$

Taking the chart norm and using $\|\Phi - \Phi'\| = |s - s'| + o(|s - s'|)$ from (27),

$$\frac{\|\Gamma_{\text{endo}}(\Phi, A) - \Gamma_{\text{endo}}(\Phi', A)\|}{\|\Phi - \Phi'\|} \longrightarrow |\lambda_A^s| \quad \text{as } s, s' \rightarrow 0, \quad s \neq s'. \quad (30)$$

By assumption (iv), the metric d is asymptotically equivalent to $\|\cdot\|$ near Φ_A^{cs} : $d(\Phi, \Phi')/\|\Phi - \Phi'\| \rightarrow 1$. Applying this to both numerator and denominator in (30) yields

$$\frac{d(\Gamma_{\text{endo}}(\Phi, A), \Gamma_{\text{endo}}(\Phi', A))}{d(\Phi, \Phi')} \longrightarrow |\lambda_A^s| \quad (31)$$

as $\Phi, \Phi' \rightarrow \Phi_A^{cs}$ in $\mathcal{M}(\Phi_0, A)$. This establishes the limit.

For the neighborhood statement, fix $\varepsilon > 0$. By the limit above, there exists $\delta_\varepsilon > 0$ such that $\Phi, \Phi' \in \mathcal{M}(\Phi_0, A)$ with $d(\Phi, \Phi_A^{cs}), d(\Phi', \Phi_A^{cs}) < \delta_\varepsilon$ imply

$$d(\Gamma_{\text{endo}}(\Phi, A), \Gamma_{\text{endo}}(\Phi', A)) \leq (|\lambda_A^s| + \varepsilon) d(\Phi, \Phi'). \quad (32)$$

Take $\mathcal{U}_A^\varepsilon := B_{\delta_\varepsilon}(\Phi_A^{cs})$, the open ball of radius δ_ε centered at the conditional steady state. ■

G Primitive conditions for Assumption 2

This appendix provides an independent route to Assumption 2 from static allocation primitives, working directly at $\mu = 1$ (rational expectations) without the homotopy of Appendix H. The key idea is that side-switching across the K -nullcline is excluded under economically interpretable bounds on households' marginal propensities to consume and sufficiently weak general-equilibrium price feedback.

Household MPC bounds. Let $c(a, z; r, w)$ denote the stationary equilibrium household consumption policy as a function of individual assets a , idiosyncratic state $z \in \mathcal{Z}$, and prices (r, w) . Along a frozen-regime continuation, aggregate consumption can change with aggregate capital because K affects prices, which in turn affects contemporaneous income components wz and ra . Current prices enter the policy only through cash-on-hand $x = wz + (1 + r)a$, so write $c(a, z; r, w) = \tilde{c}(x, z)$. A convenient “MPC bound” in this context is therefore a *Lipschitz* bound on the response of consumption to price-induced changes in cash-on-hand, rather than an MPC out of an exogenous transfer shock.

Assumption 3 (Bounded price-income MPC).

Along the frozen-regime continuation under A, there exists $\bar{m} \in (0, 1]$ such that, for all (a, z) in the support of Φ_t , all t , and all price-induced cash-on-hand levels $x' \geq x$ arising along the continuation,²

$$0 \leq \tilde{c}(x', z) - \tilde{c}(x, z) \leq \bar{m}(x' - x). \quad (33)$$

Under Assumption 3, a marginal increase in aggregate capital affects contemporaneous aggregate consumption through the induced changes in wages and returns. Define the pathwise bounds

$$\bar{Z} := \sup_{t \geq 0} \int z d\Phi_t, \quad \bar{a} := \sup_{t \geq 0} \int a d\Phi_t, \quad \mathcal{K}^{\text{seg}} := \bigcup_{t \geq 0} [\min\{K_t, K_{t+1}\}, \max\{K_t, K_{t+1}\}]. \quad (34)$$

The set $\mathcal{K}^{\text{seg}}$ is the *segment hull* of the continuation: the union of closed intervals swept out by consecutive (K_t, K_{t+1}) pairs. It contains every intermediate point at which factor-price derivatives and the nullcline slope are evaluated in the proof below. Under these bounds, the change in aggregate consumption induced by a marginal change in K is at most the aggregate MPC bound $\bar{m}$ times the induced change in contemporaneous labor and capital incomes.

²Assumption 3 bounds the response of consumption to *price-induced* changes in contemporaneous income, and differs from the standard MPC out of a transitory transfer. It is stated in increment (Lipschitz) form deliberately: with discrete idiosyncratic states, the consumption function generically has kinks—most prominently at the cash-on-hand level where the borrowing constraint begins to bind—so pointwise derivatives need not exist. No differentiability is required for the bound itself: monotonicity of consumption and of saving in cash-on-hand, standard in the income-fluctuation problem (with the concavity of [Carroll and Kimball, 1996](#) as a refinement), delivers $0 \leq \tilde{c}(x', z) - \tilde{c}(x, z) \leq x' - x$, so the conservative bound $\bar{m} = 1$ used in the Level-1 check requires neither calibration nor smoothness. Precautionary saving and borrowing constraints typically push the bound strictly below one on the relevant region of the state space; empirically, [Kaplan and Violante \(2014\)](#) estimate quarterly MPCs out of fiscal transfers in the range of 0.15–0.25 on average and up to ~ 0.4 for hand-to-mouth households. These values provide reassurance, but nothing in the verification depends on them.

Two-step verification of Assumption 2. Verification proceeds in two steps. Lemma G.1 gives a primitive sufficient condition for *endpoint no-side-switching* of the nullcline gap, i.e., $H_t H_{t+1} > 0$ for every transition t with $K_{t+1} \neq K_t$. Lemma G.2 then upgrades endpoint no-side-switching to the full line-segment no-crossing property of Assumption 2 under a curvature-margin check on the nullcline. Both steps admit direct numerical verification in a given calibration.

In the Krusell–Smith benchmark with Cobb–Douglas production, prices are smooth functions of K :

$$w_A(K) = (1 - \alpha)AK^\alpha, \quad r_A(K) = \alpha AK^{\alpha-1} - \delta, \quad (35)$$

with derivatives

$$w'_A(K) = (1 - \alpha)\alpha AK^{\alpha-1}, \quad r'_A(K) = \alpha(\alpha - 1)AK^{\alpha-2}. \quad (36)$$

Moreover, the nullcline satisfies $(C_A^{Knull})'(K) = \alpha AK^{\alpha-1} - \delta$. The next lemma gives a primitive sufficient condition for endpoint no-side-switching of the nullcline gap. It is the first step of the two-step verification.

Assumption 4 (Pathwise finite-difference bound on aggregate consumption).

Along the frozen-A continuation $\{\Phi_t\}_{t \geq 0}$, there exists $\bar{m} \in (0, 1]$ such that for every t with $K_{t+1} \neq K_t$,

$$|C_{t+1} - C_t| \leq \bar{m}(w'_A(\xi_t)\bar{Z} + |r'_A(\xi_t)|\bar{a})|K_{t+1} - K_t| \quad (37)$$

for some ξ_t between K_t and K_{t+1} , where $\bar{Z} := \sup_{t \geq 0} \int z d\Phi_t$ and $\bar{a} := \sup_{t \geq 0} \int a d\Phi_t$ are pathwise upper bounds on average productivity and average assets along the continuation.

Assumption 4 is the discrete-time analogue of the static price-income MPC bound (Assumption 3) applied to aggregate consumption along the equilibrium continuation: it caps the absolute change in C_t by the MPC bound times the maximal absolute change in contemporaneous income. The two assumptions are logically distinct, however. Assumption 3 bounds individual policy increments; Assumption 4 aggregates this into a path-level Lipschitz statement, which additionally requires that distributional composition and continuation-value effects do not push the aggregate response above the static bound. In equilibrium computations, the implied $\bar{m}$ is observable directly along the simulated continuation and falls well below the static bound for standard calibrations (e.g., ≈ 0.03 for the Krusell and Smith (1998) calibration; see Proposition G.1). The conservative envelope $\bar{m} = 1$ is tight only in the

degenerate configuration in which the entire distribution is concentrated on the constrained, slope-one branch of the consumption function—a fully hand-to-mouth economy; the gap between the envelope and the measured path coefficient reflects the small constrained mass in the ergodic distribution. A sharper bound would weight $\bar{m}$ by the constrained share, at the cost of tracking one distributional moment. Establishing Assumption 4 as a derived consequence of Assumption 3 alone is left to future work.

Lemma G.1 (Primitive condition for endpoint no-side-switching).

Fix $A \in \{B, G\}$ and consider the frozen- A continuation $\{\Phi_t\}_{t \geq 0}$ with $K_t := K(\Phi_t)$ and $C_t := C(\Phi_t, A)$. Suppose Assumption 4 holds. If, in addition,

$$\sup_{K \in \mathcal{K}^{\text{seg}}} \bar{m}(w'_A(K) \bar{Z} + |r'_A(K)| \bar{a}) - \inf_{K \in \mathcal{K}^{\text{seg}}} (C_A^{Knull})'(K) < 1, \quad (38)$$

then endpoint no-side-switching of the nullcline gap holds: $H_t H_{t+1} > 0$ for every transition t with $K_{t+1} \neq K_t$. The strict-monotonicity conclusion requires the additional condition of unique K -stationarity discussed in the main text, which rules out nonterminal flat steps with $K_{t+1} = K_t$ at $\Phi_t \neq \Phi^{cs}$.

The separated extrema in (38) are deliberate: the two mean-value points entering the proof generally differ, so the condition must bound the MPC–price term and the nullcline slope at distinct evaluation points within the segment hull. Neither the condition nor the conclusion requires K to be injective along the continuation.

Proof.

Define the nullcline gap $H_t := C_t - C_A^{Knull}(K_t)$, so that $K_{t+1} - K_t = -H_t$ along the continuation. By the mean-value theorem applied to C_A^{Knull} (a C^1 function of K under Cobb–Douglas production), there exists ξ_t^N between K_t and K_{t+1} with

$$C_A^{Knull}(K_{t+1}) - C_A^{Knull}(K_t) = (C_A^{Knull})'(\xi_t^N)(K_{t+1} - K_t). \quad (39)$$

Write $B_t := \bar{m}(w'_A(\xi_t) \bar{Z} + |r'_A(\xi_t)| \bar{a})$ and $N_t := (C_A^{Knull})'(\xi_t^N)$. The two-sided finite-difference bound (37) implies $|C_{t+1} - C_t| \leq B_t |K_{t+1} - K_t| = B_t |H_t|$. Hence

$$H_{t+1} = H_t + (C_{t+1} - C_t) - N_t(K_{t+1} - K_t) \quad (40)$$

$$= H_t + (C_{t+1} - C_t) + N_t H_t \quad (41)$$

$$= H_t(1 + N_t) + (C_{t+1} - C_t). \quad (42)$$

Bounding $|C_{t+1} - C_t| \leq B_t|H_t|$ from above and below gives

$$H_t(1 + N_t) - B_t|H_t| \leq H_{t+1} \leq H_t(1 + N_t) + B_t|H_t|. \quad (43)$$

Both mean-value points lie in the segment between K_t and K_{t+1} , hence in the segment hull $\mathcal{K}^{\text{seg}}$, so the separated extrema in the primitive condition (38) bound the difference even though $\xi_t \neq \xi_t^N$ in general: $B_t - N_t \leq \sup_{\mathcal{K}^{\text{seg}}} \bar{m}(w'_A \bar{Z} + |r'_A| \bar{a}) - \inf_{\mathcal{K}^{\text{seg}}} (C_A^{K_{\text{null}}})' < 1$, hence $1 + N_t - B_t > 0$. Sign preservation now follows by cases:

Case $H_t > 0$. Then $|H_t| = H_t$, and the lower bound gives $H_{t+1} \geq H_t(1 + N_t - B_t) > 0$.

Case $H_t < 0$. Then $|H_t| = -H_t$, and the upper bound gives $H_{t+1} \leq H_t(1 + N_t - B_t)$. Since $H_t < 0$ and $1 + N_t - B_t > 0$, the right-hand side is negative, hence $H_{t+1} < 0$.

In both cases $H_t H_{t+1} > 0$ whenever $K_{t+1} \neq K_t$, so $\{H_t\}$ cannot change sign along nondegenerate transitions. Therefore $K_{t+1} - K_t = -H_t$ has constant nonzero sign along such transitions. The strict-monotonicity conclusion further requires that $K_{t+1} = K_t$ only at $\Phi_t = \Phi^{cs}(\Phi_0, A)$ (unique K -stationarity), which rules out nonterminal flat steps. This is guaranteed by Assumption 1(i) in the representative-agent case; in the heterogeneous-agent case, it holds locally via the stable eigenvalue $\lambda_A^s \in (0, 1)$ and is maintained globally as a pathwise condition, checkable by direct inspection of the simulated continuation. ■

Lemma G.1 establishes that consecutive endpoints (K_t, C_t) and (K_{t+1}, C_{t+1}) remain on the same side of the nullcline. Because the nullcline $C = C_A^{K_{\text{null}}}(K)$ is curved, however, endpoint no-side-switching does not on its own imply that the straight line segment connecting the two points avoids the nullcline: a segment can “dip below” a concave nullcline even when both endpoints lie above it. The next lemma closes this gap by combining endpoint no-side-switching with a curvature-margin bound.

Lemma G.2 (From endpoint no-side-switching to Assumption 2).

Let $N : K \mapsto N(K)$ be C^2 on $[K_t, K_{t+1}]$ with $|N''(K)| \leq M$, and set $H_\tau := C_\tau - N(K_\tau)$ for $\tau \in \{t, t+1\}$. If $H_t H_{t+1} > 0$ and

$$\min\{|H_t|, |H_{t+1}|\} > \frac{M(K_{t+1} - K_t)^2}{8}, \quad (44)$$

then the straight line segment from (K_t, C_t) to (K_{t+1}, C_{t+1}) does not intersect the graph of N .

Proof.

Let $L(K)$ denote the line segment from (K_t, C_t) to (K_{t+1}, C_{t+1}) and define the signed gap $G(K) := L(K) - N(K)$. Then $G(K_t) = H_t$, $G(K_{t+1}) = H_{t+1}$, and $G''(K) = -N''(K)$, so

$|G''| \leq M$. Let L_H denote the linear interpolation between (K_t, H_t) and (K_{t+1}, H_{t+1}) . The standard quadratic interpolation-error bound gives

$$\sup_{K \in [K_t, K_{t+1}]} |G(K) - L_H(K)| \leq \frac{M(K_{t+1} - K_t)^2}{8}. \quad (45)$$

Since H_t and H_{t+1} have the same sign, L_H is monotone and takes values between them, with $|L_H(K)| \geq \min\{|H_t|, |H_{t+1}|\}$ on the interval. The curvature-margin condition (44) therefore yields $|G(K) - L_H(K)| < |L_H(K)|$ pointwise, which forces $G(K)$ to retain the common sign of H_t, H_{t+1} throughout. In particular $G(K) \neq 0$ on $[K_t, K_{t+1}]$, i.e. $L(K) \neq N(K)$. ■

Lemmas G.1 and G.2 together verify Assumption 2. The primitive MPC condition (38) ensures endpoint no-side-switching; the curvature-margin check (44) upgrades this to segment no-crossing. Both are directly checkable numerically along the simulated frozen-regime continuation: M is the supremum of $|(C_A^{K_{null}})''|$ on $\mathcal{K}^{\text{seg}}$, $K_{t+1} - K_t$ is the observed step size, and $|H_t|$ is the nullcline gap. In the baseline Krusell and Smith (1998) calibration, the nullcline gap $|H_t|$ —which by the resource-constraint identity equals the step size $|K_{t+1} - K_t|$ —is on the order of 10^{-2} , and M is on the order of 10^{-3} . The curvature-margin bound $M(K_{t+1} - K_t)^2/8 \approx 10^{-8}$ is therefore several orders of magnitude below $|H_t|$, and Assumption 2 holds with substantial slack.

The curvature-margin expression $M(K_{t+1} - K_t)^2/8$ has a direct economic reading. It is the product of three terms: the nullcline's local curvature M , which measures how sharply the zero-net-investment locus bends over the traversed interval; the squared one-period movement of aggregate capital $(K_{t+1} - K_t)^2$, which measures how much the economy advances per period; and the scalar $1/8$, which reflects the standard quadratic interpolation-error bound for a straight segment approximating a curved graph. The check $\min\{|H_t|, |H_{t+1}|\} > M(K_{t+1} - K_t)^2/8$ therefore says: the distance of the economy from the nullcline at the two endpoints must exceed the amount by which the nullcline bends away from its own chord over one period of capital movement. For a gently bending nullcline and small step sizes, this condition is easy to satisfy; for highly curved nullclines or large jumps, it tightens accordingly.

A local check near the conditional steady state. The global bound (38) can be weakened to a neighborhood condition near the conditional steady state. This is useful both for economic interpretation and for quantitative verification in standard calibrations.

Corollary 1 (A local check near K_A^{cs}).

Under Assumption 4, suppose there exists $\varepsilon > 0$ such that the continuation satisfies $K_t \in$

$(K_A^{cs} - \varepsilon, K_A^{cs} + \varepsilon)$ for all sufficiently large t , and

$$\sup_{|K - K_A^{cs}| \leq \varepsilon} \left(\bar{m}(w'_A(K) \bar{Z} + |r'_A(K)| \bar{a}) - (C_A^{Knull})'(K) \right) < 1. \quad (46)$$

Then there exists $T < \infty$ such that $H_t H_{t+1} > 0$ holds for all $t \geq T$ with $K_{t+1} \neq K_t$ (endpoint no-side-switching eventually). Combined with unique K -stationarity along the continuation (no nonterminal flat steps), $\{K_t\}$ is eventually strictly monotone; Assumption 2 then follows under the additional curvature-margin condition of Lemma G.2.

Proof. Restrict Lemma G.1 to the neighborhood $|K - K_A^{cs}| \leq \varepsilon$. ■

Conditions (38)–(46) are directly checkable in quantitative implementations: one evaluates the left-hand side along a simulated frozen-regime continuation using the model-implied bound $\bar{m}$ and the moments $\bar{Z}$ and $\bar{a}$. In the baseline Krusell and Smith (1998) calibration, the left-hand side of (38) evaluated along the simulated frozen- A continuation is below 0.04 even under the conservative bound $\bar{m} = 1$, well below the threshold 1, confirming that the condition holds with substantial slack.

Hierarchy of primitive sufficient conditions. The primitive condition (38) is a general statement; its usefulness in applied work depends on showing that standard macroeconomic calibrations fall inside it with slack. The next result reduces the check to a simple substitution at the conditional steady state under log utility and Cobb–Douglas production, and the subsequent numerical corollary confirms that CRRA preferences in the empirically relevant range also satisfy the primitive condition under the Krusell and Smith (1998) calibration. Together with the perfect-foresight diagnostic (Proposition 11 in the main text), these form a three-level hierarchy: an explicit pointwise bound at the conditional steady state (Level 1), a numerical extension for CRRA (Level 2), and a numerical fallback via perfect-foresight monotonicity (Level 3).

Proposition G.1 (Level 1: pointwise bound under log utility and Cobb–Douglas).

Consider a heterogeneous-household economy with log utility, Cobb–Douglas production with capital share $\alpha \in (0, 1)$ and linear depreciation $\delta \in (0, 1)$, and bounded supports for idiosyncratic states (z, a) . Define the regime-specific gross return at the conditional steady state,

$$R_A^{cs} := 1 + \alpha A (K_A^{cs})^{\alpha-1} - \delta,$$

and let $r_A^{cs} := R_A^{cs} - 1 = \alpha A (K_A^{cs})^{\alpha-1} - \delta$ denote the corresponding net return. Under Assumption 3 with $\bar{m} \leq 1$, the left-hand side of (38) evaluated at $K = K_A^{cs}$ admits the closed

form

$$\text{LHS}(K_A^{cs}) = (1 - \alpha)(r_A^{cs} + \delta)\bar{Z} + (1 - \alpha)(r_A^{cs} + \delta)\bar{a}/K_A^{cs} - r_A^{cs}, \quad (47)$$

so the primitive condition (38) holds pointwise at the conditional steady state whenever (47) is strictly below 1, and the inequality extends to a neighborhood of K_A^{cs} by continuity of $\text{LHS}(K)$.

Proof.

At K_A^{cs} , Cobb–Douglas factor prices satisfy $w'_A(K_A^{cs}) = \alpha(1 - \alpha)A(K_A^{cs})^{\alpha-1} = (1 - \alpha)(r_A^{cs} + \delta)$ and $|r'_A(K_A^{cs})| = \alpha(1 - \alpha)A(K_A^{cs})^{\alpha-2} = (1 - \alpha)(r_A^{cs} + \delta)/K_A^{cs}$, while the nullcline slope is $(C_A^{Knull})'(K_A^{cs}) = r_A^{cs}$; all three expressions use the identity $\alpha A(K_A^{cs})^{\alpha-1} = r_A^{cs} + \delta$. Substituting into (38) with $\bar{m} = 1$ yields (47). Continuity of $w'_A, r'_A, (C_A^{Knull})'$ in K and of the aggregate moments $\bar{Z}, \bar{a}$ along the continuation extends the pointwise inequality to a neighborhood of K_A^{cs} . ■

Under perfect-foresight expectations, the steady-state Euler condition pins down $R_A^{cs} = 1/\beta$ and $r_A^{cs} = 1/\beta - 1$. Under stochastic expectations over regime switches, R_A^{cs} is instead determined by the stochastic Euler $1 = \beta \mathbb{E}_A[(C_A^{cs}/C(\Phi_A^{cs}, A'))^\sigma R_A^{cs}]$ at Φ_A^{cs} , and generally differs from $1/\beta$. Formula (47) applies in either case, with R_A^{cs} evaluated at the relevant conditional steady state. Direct numerical evaluation in the Krusell and Smith (1998) calibration gives $\text{LHS}(K_A^{cs}) \approx 0.04$.

Proposition G.1 reduces Level 1 verification to direct substitution. The statement is a pointwise bound plus a continuity extension, not a global claim. The global inequality $\sup_{K \in \mathcal{K}^{\text{seg}}} \text{LHS}(K) < 1$ still requires model-specific numerical verification because $\bar{Z}$ and $\bar{a}$ depend on the realized continuation. In the standard Krusell and Smith (1998) calibration, direct evaluation gives $\text{LHS}(K_A^{cs}) \approx 0.04$, with the separated-extrema bound $\sup_{\mathcal{K}^{\text{seg}}} B_A - \inf_{\mathcal{K}^{\text{seg}}} N_A \approx 0.04$ —well below the threshold 1.

Level 2: a numerical extension to CRRA utility. Under CRRA utility with $\sigma \neq 1$, the price-income MPC bound $\bar{m}(\sigma)$ entering Assumption 3 depends on σ through the household's consumption policy. A clean analytical dependence $\bar{m}(\sigma)$ is not readily available without further structure on the idiosyncratic process. I therefore extend Level 1 to CRRA numerically: for the Krusell and Smith (1998) calibration, direct evaluation along the simulated frozen-regime continuation gives $\text{LHS}(K_A^{cs})$ below 0.2 for all $\sigma \in [0.5, 5]$, substantially below 1. This range spans the empirically relevant values of relative risk aversion. Establishing an analytical bound on the admissible σ -range from primitives remains an open question.

The hierarchy is complete: Level 1 provides an explicit pointwise bound under log utility and Cobb–Douglas. Level 2 confirms numerically that the pointwise bound also holds under CRRA in the empirically relevant range. Level 3 (Proposition 11) is a numerical fallback via perfect-foresight monotonicity, used when closed-form primitives cannot be verified. The illustration below then shows that Assumption 2 imposes a genuine geometric restriction: it can fail when the production function is non-concave.

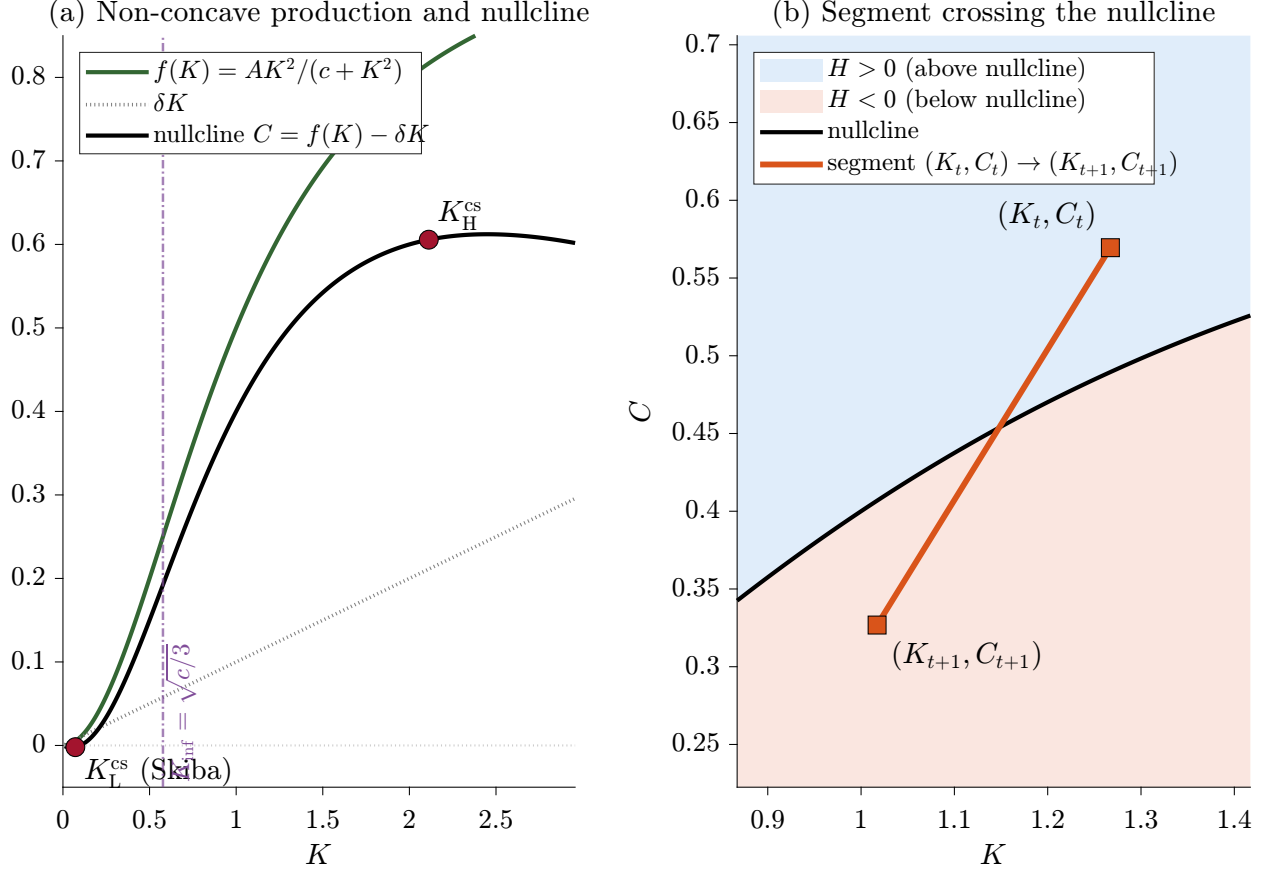
An illustration: non-concave production and the geometric possibility of segment crossing. The primitive condition rules out segment crossing under concave production and bounded MPCs. To show that Assumption 2 is a genuine geometric restriction—not automatically satisfied by the discrete-time budget constraint alone—consider a stylized economy with smooth non-concave production of the Hill form

$$f(K) = \frac{AK^2}{c + K^2}. \quad (48)$$

This f is C^∞ smooth, satisfies $f(0) = 0$ and $f'(0) = 0$, and switches from convex to concave at the inflection $K = \sqrt{c/3}$. Set $A = c = 1$, $\beta = 0.96$, $\delta = 0.10$. The steady-state Euler condition $f'(K) = 1/\beta - 1 + \delta$ then has two interior roots: a low, unstable conditional steady state $K_L^{cs} \approx 0.07$ in the convex region of f , and a standard Ramsey steady state $K_H^{cs} \approx 2.11$ in the concave region. Figure G.1(a) displays the nullcline and both steady states; the nullcline is non-monotone (rises, then falls as K increases). Panel (b) exhibits a pair of points (K_t, C_t) and (K_{t+1}, C_{t+1}) that satisfy the aggregate budget constraint $K_{t+1} = f(K_t) + (1 - \delta)K_t - C_t$ but whose one-period segment crosses the nullcline: $H_t > 0$ and $H_{t+1} < 0$, so $H_t H_{t+1} < 0$.

The illustration establishes a geometric possibility, not an actual equilibrium trajectory. Whether the optimal policy of the non-concave Ramsey problem (48) produces such trajectories would require solving for the consumption policy under the resulting non-convex value function, a substantial separate exercise. What the construction shows is narrower: non-concave technology *permits* discrete-time transitions that cross the conditional K -nullcline. Assumption 2 excludes this class of trajectories by assumption. In standard RBC and [Krusell and Smith \(1998\)](#) economies, with concave Cobb–Douglas technology and bounded household MPCs, the primitive condition (38) holds with the substantial slack reported above, and rules out any such crossings along the equilibrium path.

Figure G.1: Non-concave production: the geometric possibility of segment crossing



Notes: Panel (a): the Hill production function $f(K) = AK^2/(c + K^2)$ with $A = c = 1$, $\delta = 0.10$. The dash-dotted vertical line marks the inflection $K = \sqrt{c/3}$; f is convex to its left, concave to its right. The two conditional steady states K_L^{cs} (low, unstable) and K_H^{cs} (Ramsey) both lie on the nullcline $C = f(K) - \delta K$. Panel (b): a zoomed view of a budget-feasible segment $(K_t, C_t) \rightarrow (K_{t+1}, C_{t+1})$ crossing the nullcline. The blue half-plane marks $H > 0$ (above nullcline, $\Delta K < 0$) and the red half-plane $H < 0$ (below nullcline, $\Delta K > 0$). The segment traverses both half-planes, so $H_t H_{t+1} < 0$; Assumption 2 would exclude such a transition. The figure illustrates only that such a transition is budget-feasible under non-concave production; whether a specific equilibrium policy selects such a transition is model-specific.

Proof of Theorem 3 (within-spell K -sufficiency under log utility and Cobb–Douglas)

Proof.

Part (a): local result. Fix the conditional saddle $\mathcal{M}(\Phi_0, A)$. By the regularity conditions of Proposition 10—restated so that the orbit tail is contained in a C^1 locally invariant curve $\mathcal{W}_A$ tangent to the stable eigenspace—and the aggregate-capital projection of Appendix F, the curve $\mathcal{W}_A$ is locally homeomorphically parametrized by K in a chart neighborhood of

Φ_A^{cs} , and along it the capital deviation $k_t := K_t - K_A^{cs}$ obeys

$$k_{t+1} = \lambda_A^s k_t + o(|k_t|), \quad \lambda_A^s \in (0, 1). \quad (49)$$

Fix $\varepsilon := \frac{1}{2} \min\{\lambda_A^s, 1 - \lambda_A^s\} > 0$ and choose a chart neighborhood $\mathcal{V}_A$ of Φ_A^{cs} small enough that (i) the parametrization applies on $\mathcal{W}_A \cap \mathcal{V}_A$ and (ii) the remainder satisfies $|o(|k|)| \leq \varepsilon|k|$ there; write $\mathcal{N}_A := K(\mathcal{V}_A)$ for the K -image. The neighborhood is constructed for the fixed saddle; uniformity across initial distributions Φ_0 is not claimed. For a continuation started at $\Phi'_0 \in \mathcal{M}(\Phi_0, A) \cap \mathcal{V}_A$ with $k_t \neq 0$,

$$(\lambda_A^s - \varepsilon) k_t \leq k_{t+1} \leq (\lambda_A^s + \varepsilon) k_t \quad (k_t > 0), \quad (50)$$

and the reversed inequalities for $k_t < 0$. Since $\lambda_A^s - \varepsilon > 0$ and $\lambda_A^s + \varepsilon < 1$, three conclusions follow at once. First, k_{t+1} has the same strict sign as k_t : the path never crosses K_A^{cs} , and by induction the state remains on the parametrized piece with capital between $K(\Phi'_0)$ and K_A^{cs} , hence in $\mathcal{V}_A$. Second, $|k_{t+1}| \leq (\lambda_A^s + \varepsilon)|k_t| < |k_t|$: the sequence $\{K_t\}$ is strictly monotone and converges geometrically to K_A^{cs} . This proves (i). Third, $k_{t+1} = 0$ forces $k_t = 0$ (otherwise the lower bound gives $|k_{t+1}| \geq (\lambda_A^s - \varepsilon)|k_t| > 0$): on the parametrized piece, $\Delta K_t = 0$ only at $\Phi_t = \Phi_A^{cs}$, proving (ii). For (iii), since the parametrization is a local homeomorphism, $K_t \rightarrow K_A^{cs}$ implies $\Phi_t \rightarrow \Phi_A^{cs}$, so the distributional limit required by Theorem 1 exists; strict monotonicity of $\{K_t\}$ then permits Theorem 1 with $e(\Phi) = K(\Phi)$, yielding injectivity of K on the orbit piece in $\mathcal{V}_A$ and hence exact conditional sufficiency. No MPC bound, segment-crossing assumption, or pathwise condition has been used; the conclusions are consequences of the assumed spectral regularity.

Part (b): explicit region under Assumptions 3–4. Under log utility and Cobb–Douglas production, the factor-price derivatives and nullcline slope are explicit: $w'_A(K) = \alpha(1 - \alpha)AK^{\alpha-1}$, $|r'_A(K)| = \alpha(1 - \alpha)AK^{\alpha-2}$, and $(C_A^{Knull})'(K) = N_A(K) = \alpha AK^{\alpha-1} - \delta$, so that the functions B_A , N_A , M_A of Definition 7 (main text) match the objects of Lemma G.1, with $B_A(K) - N_A(K)$ the left-hand side of the primitive condition (38) evaluated at a common point. The endpoint cap $\bar{K} \leq (A/\delta)^{1/(1-\alpha)}$ in Definition 7 ensures $C_A^{Knull} \geq 0$ on $[\underline{K}, \bar{K}]$. Suppose hypothesis (b)(i) holds: the K -path of the continuation, together with all segments connecting consecutive capital levels, remains in the K -monotonicity interval $[\underline{K}, \bar{K}]$.

Step 1 (endpoint no-side-switching with margin). The mean-value points ξ_t (from Assumption 4) and ξ_t^N (from the nullcline expansion in Lemma G.1) lie in the segment between K_t and K_{t+1} , hence in $\mathcal{K}$. Since B_A is decreasing and N_A is decreasing on $[\underline{K}, \bar{K}]$, the endpoint

inequalities of Definition 7 bound the relevant difference at *distinct* evaluation points:

$$B_t - N_t = B_A(\xi_t) - N_A(\xi_t^N) \leq B_A(\underline{K}) - N_A(\overline{K}) \leq 1 - \kappa. \quad (51)$$

Lemma G.1 then delivers $H_t H_{t+1} > 0$ for every transition with $K_{t+1} \neq K_t$, and its sign-preservation inequality gives the quantitative version

$$|H_{t+1}| \geq (1 + N_t - B_t) |H_t| \geq \kappa |H_t|. \quad (52)$$

Step 2 (segment no-crossing). Since M_A is decreasing, the nullcline curvature on the traversed interval is bounded by $M_A(\underline{K})$; this is the constant required by Lemma G.2 regardless of the direction of the step. The step size obeys the feasibility bound $|K_{t+1} - K_t| = |H_t| = |C_t - C_A^{Knull}(K_t)| \leq C_t + C_A^{Knull}(K_t) \leq (f(K_t; A) + (1 - \delta)K_t) + f(K_t; A)$, using $0 \leq C_t \leq f(K_t; A) + (1 - \delta)K_t$ (resource feasibility) and $0 \leq C_A^{Knull}(K) \leq f(K; A)$ on $[\underline{K}, \overline{K}]$. Since $2f(K; A) + (1 - \delta)K$ is increasing, the second endpoint inequality of Definition 7 gives $|H_t| \leq 2f(\overline{K}; A) + (1 - \delta)\overline{K} < 8\kappa/M_A(\underline{K})$. The curvature-margin condition of Lemma G.2 requires $\min\{|H_t|, |H_{t+1}|\} > M_A(\underline{K}) (K_{t+1} - K_t)^2/8 = (M_A(\underline{K}) |H_t|/8) |H_t|$. For the first argument of the minimum, $M_A(\underline{K}) |H_t|/8 < \kappa \leq 1$ gives the strict inequality directly. For the second, Step 1 gives $|H_{t+1}| \geq \kappa |H_t| > (M_A(\underline{K}) |H_t|/8) |H_t|$, again using $M_A(\underline{K}) |H_t|/8 < \kappa$. Steps 1 and 2 together verify Assumption 2 along the continuation, and the sign-preservation of H_t makes $\{K_t\}$ strictly monotone across nondegenerate transitions.

Step 3 (sufficiency under hypothesis (b)(ii)). Two cases. If no flat step occurs ($K_{t+1} \neq K_t$ for all t), then by Steps 1–2 the sequence $\{K_t\}$ is strictly monotone, and it is bounded (by feasibility, $K_{t+1} \leq f(K_t; A) + (1 - \delta)K_t$, and $K_t \geq 0$), hence convergent; the limit state is K -stationary with capital in $\mathcal{K}$, so by hypothesis (b)(ii) it lies in the chart neighborhood $\mathcal{V}_A$, where part (a)(ii) identifies it with Φ_A^{cs} . If a flat step occurs at some finite t ($K_{t+1} = K_t$), the state Φ_t is K -stationary with capital in $\mathcal{K}$, so by hypothesis (b)(ii) $\Phi_t \in \mathcal{V}_A$, and part (a)(ii) forces $\Phi_t = \Phi_A^{cs}$: the flat step is terminal. In both cases $K_t \rightarrow K_A^{cs}$ strictly monotonically before absorption, and Theorem 1 with $e = K$ delivers exact conditional sufficiency. ■

Numerical verification under the Krusell and Smith (1998) calibration (Section 4.2 of the main text). With $\alpha = 0.36$, $\delta = 0.025$, $A \in \{0.99, 1.01\}$, $\bar{Z} \approx 1$, $\bar{a} \approx \sup_t K_t \approx 38.5$, and $\bar{m} \leq 1$, take $[\underline{K}, \overline{K}] = [34.6, 38.5]$, the ergodic range. The endpoint evaluations are $B_A(\underline{K}) \approx 0.050$ and $N_A(\overline{K}) \approx 0.010$, so $B_A(\underline{K}) - N_A(\overline{K}) \approx 0.040 < 0.05 = 1 - \kappa$ with $\kappa = 0.95$. For the curvature inequality, $M_A(\underline{K}) \approx 6.9 \times 10^{-4}$, so $8\kappa/M_A(\underline{K}) \approx 1.1 \times 10^4$, while $2f(\overline{K}; A) + (1 - \delta)\overline{K} \approx 2(3.7) + 37.5 \approx 45$ —a margin exceeding two orders of magnitude. The productivity bound $\bar{Z} \leq z_{\max}$ is a priori from compact support; $\bar{a}$ and the

ergodic range are read off the computed equilibrium, so the check validates the computed continuation rather than replacing computation. Replacing $\bar{a}$ by the a priori feasibility cap $(A/\delta)^{1/(1-\alpha)} \approx 319$ yields $B_A(\underline{K}) - N_A(\bar{K}) \approx 0.23 < 1$, so the inequalities can still be verified from calibrated parameters alone.

Why the residual condition resists primitive closure. The single hypothesis left unproved in Theorem 3(b) is that the continuation admits no K -stationary state with capital in $\mathcal{K}$ outside the chart neighborhood $\mathcal{V}_A$. Closing it from primitives requires characterizing the basin of attraction of the conditional steady state for the distributional dynamics: one must rule out distributional configurations at which aggregate net saving vanishes while the distribution itself keeps evolving (a moving plateau)—including configurations whose *capital* happens to lie near K_A^{cs} while the distribution is far from Φ_A^{cs} , which is why the condition must be stated in state space rather than in capital units. The MPC apparatus of this appendix controls the *change* in aggregate consumption against the *change* in aggregate capital, and is silent when both are zero. A primitive treatment would need either a strict-monotonicity property of aggregate saving in a distributional ordering (a stochastic-dominance argument) or a global Lyapunov function for $\Gamma_{\text{endo}}(\cdot, A)$; both appear to require machinery beyond the scope of this paper. The condition is directly checkable along any computed continuation, and fails to bind in every calibration examined here.

H Continuous dependence on beliefs

This appendix establishes two distinct continuity results used by the PF diagnostic (Proposition 11): (a) continuity of the finite-horizon frozen-regime continuation $\{\Phi_t(\mu)\}_{t=0}^T$ from a given initial Φ_0 , used to transport endpoint no-side-switching from $\mu = 0$ to $\mu = 1$; and (b) continuity of the conditional steady state $\Phi_A^{cs}(\mu)$, used to anchor the limit of the continuation. Neither result requires the ergodic/stationary distribution of the full stochastic process.

Proof of continuous dependence on μ .

The argument proceeds in four steps.

Step 1 (Continuity of household policies). For given prices (r, w) and belief parameter $\mu \in [0, 1]$, the household's recursive problem is

$$v(a, z; \mu) = \max_{c, a'} u(c) + \beta \sum_{A'} \Pi(\mu)(A'|A) \mathbb{E}_{z'|z} v(a', z'; \mu),$$

subject to the budget constraint and borrowing limit. The natural borrowing limit pins down

the lower bound $\underline{a}$, while the upper bound $\bar{a}$ is the maintained boundary of a compact asset interval $[\underline{a}, \bar{a}]$ on which the analysis is conducted—this is the standard numerical convention in Aiyagari/Krusell–Smith implementations and is supported, for example, by precautionary-saving arguments combined with concavity that put a uniform tightness bound on the support of stationary distributions when prices lie in a compact set. On this compact asset domain, and under strict concavity and Inada conditions, the Bellman operator T_μ is a contraction in the sup-norm for each μ (Stokey et al., 1989, Theorem 9.6). Since $\Pi(\mu) = (1 - \mu)I + \mu\Pi$ enters linearly in the expectation operator, T_μ is continuous in μ , so its unique fixed point $v(\cdot; \mu)$ depends continuously on μ by the parametric contraction mapping theorem. Continuity of the policy functions $c(\cdot; \mu)$ and $a'(\cdot; \mu)$ follows by the theorem of the maximum.

Step 2 (Continuity of the frozen-regime operator). Fix a regime A . The frozen-regime endogenous law of motion $\Gamma_{\text{endo}}(\Phi, A; \mu)$ maps a distributional state Φ to the pushforward of Φ under one period of the individual savings policy $a'(\cdot; \mu)$ at prices $(r(\Phi, A), w(\Phi, A))$. As a composition of (i) the continuous price map $\Phi \mapsto (r(\Phi, A), w(\Phi, A))$, (ii) the jointly continuous individual policy $a'(\cdot; \mu)$ from Step 1, and (iii) the pushforward operation, $\Gamma_{\text{endo}}(\cdot, A; \mu)$ is jointly continuous in (Φ, μ) in the product topology (weak convergence on distributions, Euclidean on μ).

Step 3 (Continuity of finite iterates). Fix an initial distribution Φ_0 and a horizon T . The t -fold iterate $\Gamma_{\text{endo}}^t(\Phi_0, A; \mu)$ is the composition of t jointly continuous maps from Step 2, each continuous in μ . Hence the entire frozen-regime continuation $\{\Phi_t(\mu)\}_{t=0}^T$, and therefore the aggregate sequence $\{(K_t(\mu), C_t(\mu))\}_{t=0}^T$, depends continuously on μ pointwise in t for every finite T .

Step 4 (Continuity of the conditional steady state). The conditional steady state $\Phi_A^{cs}(\mu)$ satisfies the fixed-point equation $\Phi_A^{cs}(\mu) = \Gamma_{\text{endo}}(\Phi_A^{cs}(\mu), A; \mu)$. Fix $\mu^* \in [0, 1]$ and assume that at μ^* the regularity conditions of Proposition 10 hold: $\Gamma_{\text{endo}}(\cdot, A; \mu^*)$ is Fréchet-differentiable at $\Phi_A^{cs}(\mu^*)$, the stable eigenvalue $\lambda_A^{cs}(\mu^*) \in (0, 1)$ is simple and real, and no other eigenvalue of the derivative sits on the unit circle. Then the operator $I - D\Gamma_A(\mu^*)$ is invertible on the tangent space at $\Phi_A^{cs}(\mu^*)$. Applying the implicit function theorem to $F(\Phi, \mu) := \Phi - \Gamma_{\text{endo}}(\Phi, A; \mu) = 0$, and using joint continuity of Γ_{endo} in (Φ, μ) from Step 2 together with continuity of the derivative in μ , yields a unique $\Phi_A^{cs}(\mu)$ in a neighborhood of μ^* that varies continuously in μ . ■

I Details for the openness step in Proposition 11 (perfect-foresight diagnostic)

This appendix provides the tail argument used in the openness step of Proposition 11.

Tail monotonicity via eigenvalue continuity.

Fix $\mu^* \in \mathcal{S}$. Strict monotone convergence of $\{K_t(\mu^*)\}$ to $K_A^{cs}(\mu^*)$ implies that, near the conditional steady state, the linearized frozen-regime dynamics

$$K_{t+1} - K_A^{cs}(\mu^*) \approx \lambda_s(\mu^*) (K_t - K_A^{cs}(\mu^*)) \quad (53)$$

have a stable eigenvalue $\lambda_s(\mu^*) \in (0, 1)$. If $\lambda_s(\mu^*)$ were negative or complex, convergence near the steady state would be oscillatory, contradicting monotonicity of $\{K_t(\mu^*)\}$ in a neighborhood of its limit.

The eigenvalue $\lambda_s(\mu)$ is the stable root of the linearized Euler equation–resource constraint system at $K_A^{cs}(\mu)$. Both the steady state $K_A^{cs}(\mu)$ and the coefficients of the linearized system depend continuously on μ (by the continuous dependence established in Appendix H). Since eigenvalues of a matrix depend continuously on its entries, $\lambda_s(\mu)$ is continuous in μ .

Therefore, there exists $\varepsilon_1 > 0$ such that $|\mu - \mu^*| < \varepsilon_1$ implies $\lambda_s(\mu) \in (0, 1)$, ensuring monotone convergence near $K_A^{cs}(\mu)$.

Choose T large enough that $K_t(\mu^*)$ lies within a neighborhood of $K_A^{cs}(\mu^*)$ where the local map is monotone with slope in $(0, 1)$, hence order-preserving. By continuity of $(K_t(\mu), C_t(\mu))$ in μ for each finite t , for μ sufficiently close to μ^* , the path $\{K_t(\mu)\}_{t \geq T}$ also enters this neighborhood and inherits monotone convergence from the eigenvalue condition. Combined with the finite-horizon continuity argument (which preserves the sign of H_t for $t \leq T$), monotonicity holds for all t , proving $\mu \in \mathcal{S}$. ■

J Endogenous labor supply extensions

This appendix provides the derivations for the endogenous labor supply extensions discussed in Section 4.3.

GHH preferences. Consider preferences $u(c, l_H) = \frac{1}{1-\sigma}(c - \frac{\eta}{1+1/\chi} l_H^{1+1/\chi})^{1-\sigma}$ with budget constraint $c + a' = a(1 + r(\Phi, A)) + w(\Phi, A)z l_H$, where z denotes idiosyncratic labor

productivity. The individual labor-supply optimality condition yields:

$$l_H(a, z; \Phi, A) = \left(\frac{z}{\eta}\right)^\chi w(\Phi, A)^\chi, \quad L(\Phi, A) = w(\Phi, A)^\chi \int \frac{z^{\chi+1}}{\eta^\chi} d\Phi_z, \quad (54)$$

where Φ_z is the marginal distribution of productivity. Defining $M = \int \frac{z^{\chi+1}}{\eta^\chi} d\Phi_z$, the labor market clearing condition $(1 - \alpha)AK^\alpha L^{-\alpha} = w = (L/M)^{1/\chi}$ implies:

$$c_A^{Knull} = A \left(M^{\frac{1}{1+\alpha\chi}} ((1 - \alpha)A)^{\frac{\chi}{1+\alpha\chi}} \right)^{1-\alpha} (K_A^{Knull})^{\alpha+(1-\alpha)\frac{\alpha\chi}{1+\alpha\chi}} - \delta K_A^{Knull}. \quad (55)$$

Since M depends only on the stationary productivity distribution (not the wealth distribution), the nullcline is distribution-free and the monotonicity argument applies.

CRRA preferences. Under $u(c, l_H) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{\eta}{1+1/\chi} l_H^{1+1/\chi}$, the individual labor supply becomes:

$$l_H(a, z; \Phi, A) = \left(\frac{z}{\eta c(a, z; \Phi, A)^\sigma} \right)^\chi w(\Phi, A)^\chi. \quad (56)$$

Following the same derivation, the conditional K -nullcline is:

$$c_A^{Knull} = A \left(M(\Phi^{Knull})^{\frac{1}{1+\alpha\chi}} ((1 - \alpha)A)^{\frac{\chi}{1+\alpha\chi}} \right)^{1-\alpha} (K_A^{Knull})^{\alpha+(1-\alpha)\frac{\alpha\chi}{1+\alpha\chi}} - \delta K_A^{Knull}, \quad (57)$$

where $M(\Phi^{Knull}) = \int \frac{z^{\chi+1}}{\eta^\chi c^\sigma} d\Phi^{Knull}$ now depends on the wealth distribution through individual consumption. This distributional dependence prevents analytical verification of monotonicity.

K Search and matching model

This appendix provides the full specification of the Diamond–Mortensen–Pissarides economy used in Section 5. The model is a discrete-time stochastic version of the canonical search-and-matching framework with exogenous separation, Nash-bargained wages, and Cobb–Douglas matching, solved at monthly frequency.

Environment. A representative household comprises a unit measure of homogeneous workers. Employed workers produce output at productivity z and earn a bilaterally determined wage w . Unemployed workers engage in home production with flow value $b > 0$. Firms post vacancies at per-unit cost $\kappa > 0$.

Aggregate states. The aggregate state is

$$S = [u_{-1}, z], \quad (58)$$

where $u_{-1} = 1 - n_{-1}$ is the inherited unemployment rate (the predetermined endogenous state) and z is aggregate TFP. Log productivity follows an AR(1) process,

$$\ln z_t = \rho_z \ln z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_z^2), \quad (59)$$

discretized into an 11-state Markov chain using the Tauchen method with ± 3 standard deviation bounds.

Matching. A Cobb–Douglas matching function governs the flow of new hires:

$$M(u, v) = m_0 u^\xi v^{1-\xi}, \quad (60)$$

where $u = 1 - n_{-1}$ is the unemployment level, v is aggregate vacancies, $m_0 > 0$ is matching efficiency, and $\xi \in (0, 1)$ is the matching elasticity. Labor market tightness is defined as $\theta := v/u$. The vacancy-filling probability is

$$q(\theta) = \frac{M(u, v)}{v} = m_0 \theta^{-\xi}, \quad (61)$$

and the job-finding probability is $f(\theta) = m_0 \theta^{1-\xi}$. Because the Cobb–Douglas specification can imply $q(\theta) > 1$ for low θ , I truncate q at unity in the numerical implementation.

Employment dynamics. Employment evolves according to

$$n = (1 - \lambda) n_{-1} + q(\theta) v, \quad (62)$$

where $\lambda \in (0, 1)$ is the exogenous job separation rate.

Resource constraint. Aggregate consumption equals market output net of vacancy posting costs plus home production by the unemployed:

$$c(S) = n(S) z - \kappa v(S) + (1 - n(S)) b. \quad (63)$$

Household preferences. The household has temporal CRRA utility with risk aversion $\sigma > 0$ and discount factor $\beta \in (0, 1)$. The stochastic discount factor between aggregate states

S and S' is

$$\Lambda(S, S') = \beta \left(\frac{c(S)}{c(S')} \right)^\sigma. \quad (64)$$

Wage determination. Wages are determined by Nash bargaining with worker bargaining weight $\eta \in (0, 1)$:

$$w(S) = (1 - \eta)b + \eta(z + \kappa\theta(S)). \quad (65)$$

The first term reflects the worker's outside option (home production), and the second captures the worker's share of the match surplus inclusive of the firm's saved vacancy costs.

Free entry. Firms post vacancies until the expected value of a filled vacancy equals the posting cost. The free-entry condition is

$$\frac{\kappa}{q(S)} = \beta \mathbb{E} \left[\left(\frac{c(S)}{c(S')} \right)^\sigma \left(z' - w(S') + (1 - \lambda) \frac{\kappa}{q(S')} \right) \middle| S \right]. \quad (66)$$

The left-hand side is the expected cost of filling a vacancy. The right-hand side is the discounted value of a filled job. The matched worker produces next period, earning flow profit $z' - w(S')$. Separation then occurs with probability λ , so the continuation value $\kappa/q(S')$ is weighted by the survival probability $(1 - \lambda)$.

Recursive competitive equilibrium. A recursive competitive equilibrium consists of policy functions $\{\theta(S), v(S), n(S), w(S), c(S)\}$ such that:

1. The free-entry condition (66) holds for all S .
2. Wages satisfy the Nash bargaining outcome (65).
3. Employment evolves according to (62).
4. The resource constraint (63) holds.
5. Expectations are consistent with the induced law of motion for S .

The model is solved globally using the repeated transition method (Lee, 2025).

Calibration. Table K.1 reports the parameter values. The model is calibrated at monthly frequency. The discount factor $\beta = 0.99^{1/3} \approx 0.9967$ corresponds to a quarterly discount factor of 0.99. The matching elasticity $\xi = 0.635$ falls within the range surveyed by Petrongolo and Pissarides (2001). The worker's bargaining weight satisfies the Hosios condition $\eta = \xi$, ensuring constrained efficiency of the decentralized equilibrium absent aggregate risk. The

separation rate $\lambda = 0.028$ per month is consistent with JOLTS data. Log TFP follows an AR(1) with persistence $\rho_z = 0.950$ and innovation standard deviation $\sigma_z = 0.050$, discretized into 11 states via the Tauchen method with ± 3 standard deviation bounds.

Table K.1: DMP model: calibrated parameters

Parameter	Symbol	Value	Target / Source
Discount factor	β	$0.99^{1/3} \approx 0.9967$	Quarterly $\beta = 0.99$
Risk aversion	σ	1.000	Log utility
Matching efficiency	m_0	0.543	Steady-state job-finding rate
Matching elasticity	ξ	0.635	Petrongolo and Pissarides (2001)
Separation rate	λ	0.028	JOLTS monthly separation rate
Vacancy posting cost	κ	0.309	Steady-state vacancy rate
Home production	b	0.400	
Bargaining weight	η	0.635	Hosios condition ($\eta = \xi$)
TFP persistence	ρ_z	0.950	Monthly AR(1)
TFP innovation s.d.	σ_z	0.050	Output volatility
TFP grid points		11	Tauchen, ± 3 s.d.

Benefit extensions. Two extensions of the baseline benefit structure are considered here (the main text uses only constant b). The first extension introduces independently varying stochastic benefits $b \in \{b_L, b_H\}$ following a two-state Markov chain with persistence $\rho_b = 0.98$. The low benefit equals the baseline value $b_L = b = 0.400$, and the high benefit is $b_H = b + 0.50 w^{ss}$, where w^{ss} is the steady-state wage. The exogenous state expands to $(z, b) \in \{B, G\} \times \{b_L, b_H\}$ with a Kronecker-product transition matrix, producing four conditional saddle paths. The slope ordering is (G, b_L) steepest, (B, b_H) flattest, separating into two productivity groups each split by the benefit level.

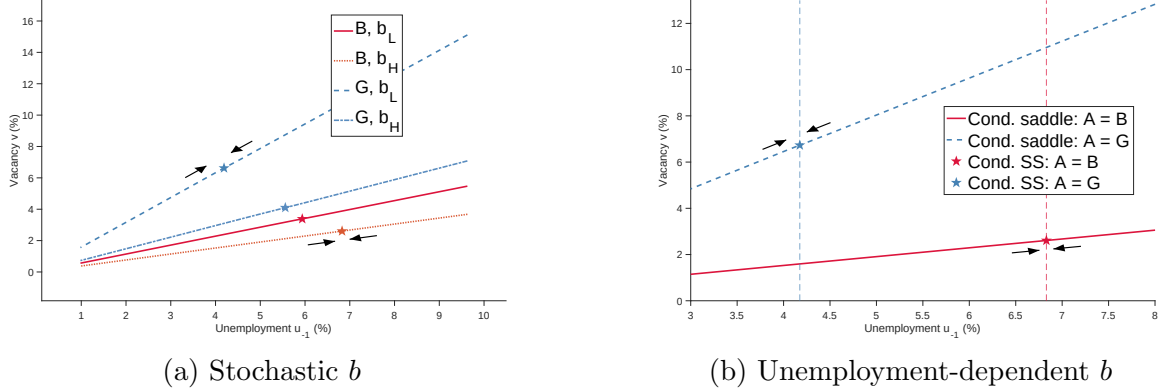
The second extension implements unemployment-dependent benefits as an automatic stabilizer. The benefit level is

$$b(u) = b + \omega(u) \bar{b} w^{ss}, \quad \omega(u) = \min\left\{1, \max\left\{0, \frac{u - u_\ell}{u_h - u_\ell}\right\}\right\},$$

where $u_\ell = 0.055$ and $u_h = 0.070$ are unemployment thresholds and $\bar{b} = 1.50$ scales the top-up relative to the steady-state wage w^{ss} . Benefits equal the baseline $b = 0.400$ when $u \leq u_\ell$ and rise linearly to $b + \bar{b} w^{ss}$ when $u \geq u_h$. Because benefits depend on unemployment rather than forming a separate exogenous state, the model has two conditional saddle paths indexed by TFP alone. Along each saddle, benefits rise as unemployment increases; the effect is asymmetric because the B regime settles at higher unemployment, triggering higher

benefits and compressing the surplus, while the G saddle retains meaningful slope. The slope asymmetry and steady-state unemployment gap both widen relative to the constant-benefit baseline.

Figure K.1: Conditional saddle paths under alternative benefit regimes



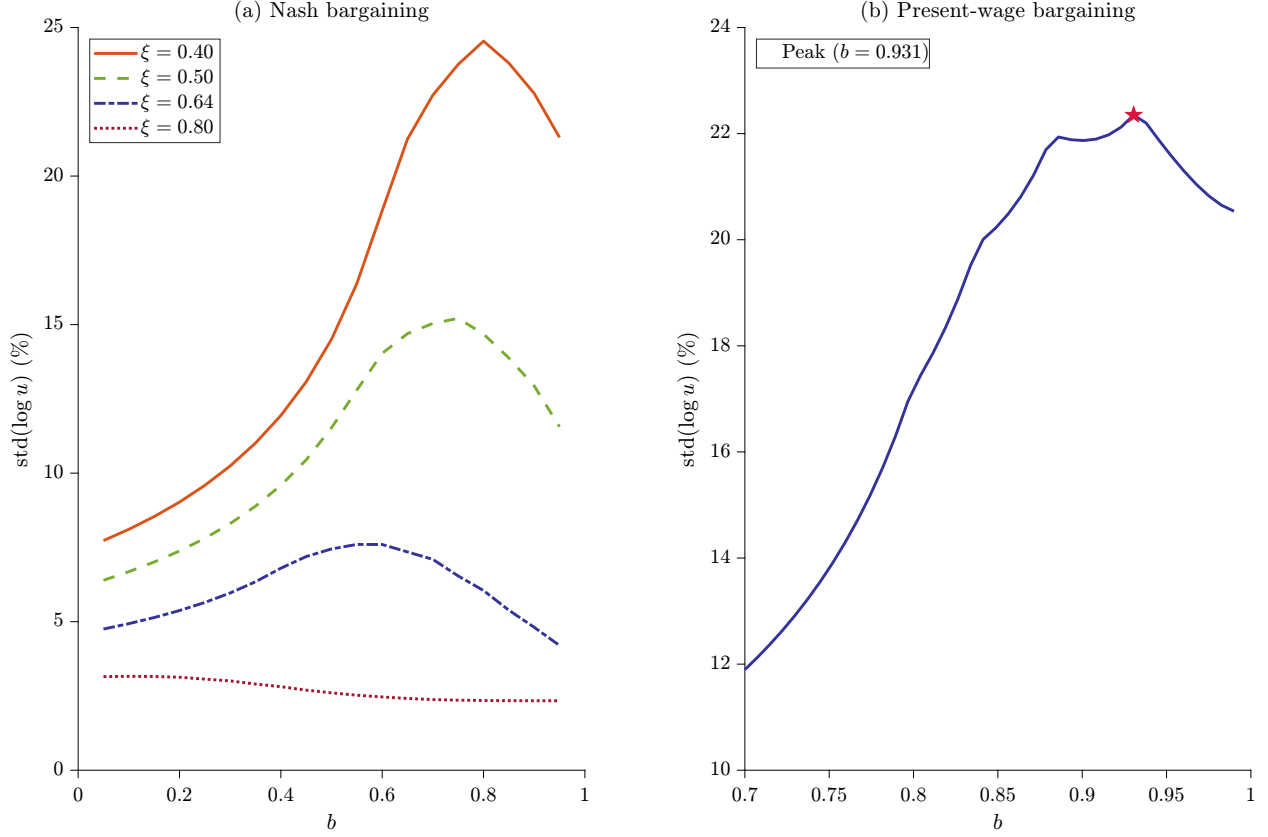
Notes: Panel (a): independently stochastic benefits $b \in \{b_L, b_H\}$ yield four conditional saddles, (G, b_L) steepest and (B, b_H) flattest. Panel (b): benefits depend on unemployment (linearly increasing in u above a threshold), so the exogenous state remains two TFP levels. Stars mark conditional steady states.

Sensitivity of the hump-shaped volatility result. Figure K.2 reports the robustness of the non-monotone relationship between benefit generosity and unemployment volatility documented in Section 5. Panel (a) varies the matching elasticity $\xi \in \{0.40, 0.50, 0.635, 0.80\}$ under Nash bargaining, holding all other parameters at baseline values. Panel (b) replaces Nash bargaining with present-wage bargaining, $w = (1 - \eta)b + \eta z$, which removes the congestion externality feedback from tightness to wages. The hump shape persists across wage-setting rules and for matching elasticities up to the baseline: under Nash bargaining, the peak shifts leftward as ξ increases, flattening entirely at $\xi = 0.80$; under present-wage bargaining, the peak occurs at $b \approx 0.93$. The non-monotone pattern is therefore driven by surplus geometry rather than a specific wage-feedback channel.

L Continuous-time RBC benchmark

This appendix formulates the continuous-time analogue of the representative-agent RBC economy of Section 2. In continuous time, the conditional saddle path is a one-dimensional invariant manifold on which capital converges strictly monotonically to the conditional steady state. The key difference from discrete time is that monotonicity is automatic: no analogue

Figure K.2: Robustness of hump-shaped volatility: matching elasticity (panel a) and wage-setting (panel b)



Notes: Panel (a): $\text{std}(\log u)$ as a function of b under Nash bargaining for matching elasticities $\xi \in \{0.40, 0.50, 0.635, 0.80\}$. The hump shape persists for $\xi \leq 0.635$; at $\xi = 0.80$, volatility is nearly flat. Panel (b): $\text{std}(\log u)$ under present-wage bargaining ($w = (1 - \eta)b + \eta z$). The peak occurs at $b \approx 0.93$.

of Assumption 2 (the segment-crossing condition) is needed, because the continuity of the equilibrium trajectory prevents capital from jumping across the K -nullcline.

Environment. Time is continuous, $t \in [0, \infty)$. A representative household with discount rate $\rho > 0$ maximizes

$$\max \mathbb{E}_0 \int_0^\infty e^{-\rho t} \log C_t dt. \quad (67)$$

Aggregate TFP $A_t \in \{B, G\}$ with $G > B > 0$ follows a continuous-time two-state Markov chain with Poisson transition intensities $\lambda_{BG} > 0$ (from B to G) and $\lambda_{GB} > 0$ (from G to B). Production is Cobb–Douglas with capital share $\alpha \in (0, 1)$ and depreciation rate $\delta > 0$.

Labor supply is inelastic and normalized to unity. The resource constraint is

$$\dot{K} = A_t K^\alpha - \delta K - C, \quad (68)$$

and competitive factor prices are

$$r(K, A) = A\alpha K^{\alpha-1} - \delta, \quad w(K, A) = A(1 - \alpha)K^\alpha. \quad (69)$$

HJB equation. Let $V(K, A)$ denote the value function of the representative household. Write λ_A for the intensity of leaving regime A (so $\lambda_B = \lambda_{BG}$ and $\lambda_G = \lambda_{GB}$), and let $A' \neq A$ denote the other regime. The Hamilton–Jacobi–Bellman equation is

$$\rho V(K, A) = \max_C \left\{ \log C + V_K(K, A)[AK^\alpha - \delta K - C] + \lambda_A[V(K, A') - V(K, A)] \right\}. \quad (70)$$

Optimality conditions. The first-order condition for consumption is

$$\frac{1}{C} = V_K(K, A), \quad \text{so} \quad C(K, A) = \frac{1}{V_K(K, A)}. \quad (71)$$

Differentiating the HJB with respect to K and substituting $V_K = 1/C$ yields the consumption Euler equation:

$$\frac{\dot{C}}{C} = r(K, A) - \rho + \lambda_A \left[\frac{C(K, A)}{C(K, A')} - 1 \right]. \quad (72)$$

The first two terms are the standard Keynes–Ramsey rule: consumption growth equals the gap between the return on capital and the discount rate. The third term captures the anticipation of regime switching. If a transition to A' would lower consumption ($C(K, A') < C(K, A)$), the household tilts consumption toward the future, raising $\dot{C}/C$.

Conditional saddle path. Fix a regime $A \in \{B, G\}$ and an initial capital stock $K_0 > 0$. If we freeze the exogenous state at A while retaining the stochastic-equilibrium consumption policy $C(K, A)$, the economy evolves according to an autonomous ordinary differential equation:

$$\dot{K} = F_A(K) := AK^\alpha - \delta K - C(K, A). \quad (73)$$

The conditional saddle path under A is the trajectory $\{K(t)\}_{t \geq 0}$ solving (73) from K_0 . The conditional K -nullcline,

$$C_A^{Knull}(K) = AK^\alpha - \delta K, \quad (74)$$

is identical to its discrete-time counterpart (Proposition 1). The conditional steady state K_A^{cs} is the unique capital level at which $F_A(K_A^{cs}) = 0$ —equivalently, the point where the consumption policy meets the K -nullcline: $C(K_A^{cs}, A) = C_A^{Knull}(K_A^{cs})$.

Monotonicity without Assumption 2.

Proposition L.1 (Monotone convergence in continuous time).

Suppose the stochastic equilibrium is such that the consumption policy $C(\cdot, A)$ is continuous and, for each $A \in \{B, G\}$:

- (i) *there exists a unique conditional steady state K_A^{cs} with $F_A(K_A^{cs}) = 0$; and*
- (ii) *for any $K_0 > 0$, the frozen-regime trajectory satisfies $K(t) \rightarrow K_A^{cs}$ as $t \rightarrow \infty$.*

Then capital converges to K_A^{cs} strictly monotonically: $\dot{K}(t) > 0$ for all t if $K_0 < K_A^{cs}$, and $\dot{K}(t) < 0$ for all t if $K_0 > K_A^{cs}$. In particular, no analogue of Assumption 2 is needed.

Proof.

Step 1 (Sign persistence of F_A). By hypothesis (i), $F_A(K_A^{cs}) = 0$ and K_A^{cs} is the unique zero of F_A on $\mathbb{R}_{++}$. The function F_A is continuous, being the difference of the C^1 net production function $AK^\alpha - \delta K$ and the continuous equilibrium consumption policy $C(\cdot, A)$. By the intermediate value theorem, F_A cannot change sign on $(0, K_A^{cs})$ or on (K_A^{cs}, ∞) without possessing another zero.

Step 2 (Direction from stability). Hypothesis (ii) requires $K(t) \rightarrow K_A^{cs}$. If $K_0 < K_A^{cs}$, convergence forces $\dot{K} = F_A(K) > 0$ in a neighborhood of K_A^{cs} . By Step 1, $F_A(K) > 0$ for all $K \in (0, K_A^{cs})$. The symmetric argument applies when $K_0 > K_A^{cs}$.

Step 3 (Strict monotonicity). Because $\dot{K} = F_A(K)$ is nonzero along the trajectory for $K \neq K_A^{cs}$, the path $t \mapsto K(t)$ is strictly monotone. ■

Remark. The key difference from the discrete-time monotonicity result (Proposition 2) is that a continuous trajectory cannot “jump” across the K -nullcline. In discrete time, the finite step $\Delta K_t = K_{t+1} - K_t$ can carry the state past the nullcline in a single period, necessitating Assumption 2 to exclude such overshooting. In continuous time, the flow $\dot{K} = F_A(K)$ must

pass through zero before changing sign, and the unique-zero condition prevents this. Thus, the one-dimensional topology of the conditional saddle, combined with the continuity of the flow, delivers monotonicity as a purely topological consequence of stability and uniqueness.

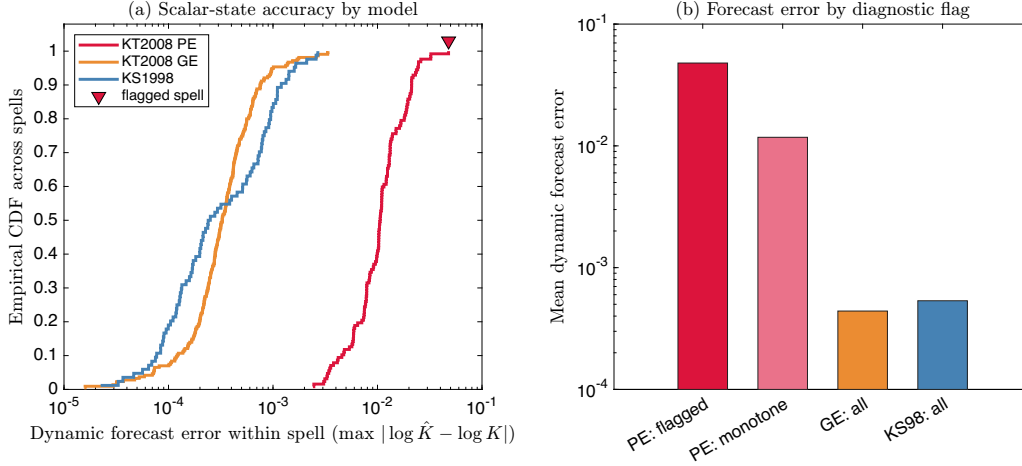
M Diagnostic validation in a lumpy-investment economy

This appendix carries out the boundary test summarized in Section 4: in the lumpy-investment economy of [Khan and Thomas \(2008\)](#), where nullcline invariance fails and K -sufficiency is no longer a theorem, does the within-spell monotonicity diagnostic track scalar-state forecast accuracy?

Design. The exercise compares three economies solved with the same global method: the [Khan and Thomas \(2008\)](#) model in general equilibrium, the same model with prices fixed at their stationary-equilibrium values (the partial-equilibrium experiment under which [Khan and Thomas](#) themselves show lumpiness matters), and the [Krusell and Smith \(1998\)](#) benchmark. For each iso-regime spell of the simulated path (length at least three), two statistics are computed. The first is the monotonicity diagnostic $|\sum_t \Delta K_t| / \sum_t |\Delta K_t|$, equal to one if and only if capital moves monotonically within the spell, as Proposition 8 predicts where its hypotheses hold; spells containing a sign change of ΔK are *flagged*. The second is the scalar-state forecast error: a regime-specific log-linear rule $\log K' = b_0(A) + b_1(A) \log K$ is fitted to the full sample and iterated dynamically within the spell from the realized initial capital—the multi-period standard of [Den Haan \(2010\)](#)—and the maximum log-deviation from the realized path is recorded.

Results. Figure [M.1](#) validates both sides of the prediction. On the success side, every one of the 215 spells of the [Khan and Thomas \(2008\)](#) general-equilibrium economy and all 84 spells of the [Krusell and Smith \(1998\)](#) benchmark are exactly monotone in K ; dynamic forecast errors average 4×10^{-4} in log terms. The diagnostic thus *rationalizes* the finding of [Khan and Thomas \(2008\)](#) that approximate aggregation survives lumpy investment: general-equilibrium price movements smooth aggregate investment enough that the within-spell capital path remains monotone, and where the monotonicity holds, scalar-state forecasting is accurate. On the failure side, removing the price feedback breaks the monotonicity and the accuracy together: in the partial-equilibrium economy, interior monotonicity values and flagged spells appear, mean dynamic errors rise by a factor of roughly thirty (to 1.2×10^{-2}), and flagged spells carry errors roughly four times larger than monotone spells of comparable

Figure M.1: The monotonicity diagnostic tracks scalar-state accuracy



Notes: Panel (a): empirical CDF of within-spell dynamic forecast errors (maximum $|\log \hat{K} - \log K|$ from iterating the fitted regime-specific log-linear rule) across iso-regime spells of length at least three, for the [Khan and Thomas \(2008\)](#) model in partial equilibrium (crimson), in general equilibrium (orange), and the [Krusell and Smith \(1998\)](#) benchmark (steel blue). Triangles mark spells flagged by the monotonicity diagnostic (a sign change of ΔK within the spell). Panel (b): mean dynamic forecast error by group. In the general-equilibrium economies every spell is exactly monotone and no spell is flagged.

length. Notably, the one-step R^2 of the fitted rule remains 0.998 in the partial-equilibrium economy—high enough to look reassuring by the conventional metric—while the dynamic errors reveal the strain, echoing the critique of [Den Haan \(2010\)](#). The diagnostic is computable along any simulated path at negligible cost; where it is clean, scalar-state methods are on solid ground, and where it degrades, the cross-sectional distribution carries independent forecasting power, consistent with the evidence in [Bachmann et al. \(2013\)](#).

References

- AIYAGARI, S. R. (1994): “Uninsured Idiosyncratic Risk and Aggregate Saving,” *The Quarterly Journal of Economics*, 109, 659–684, publisher: Oxford University Press.
- BACHMANN, R., R. J. CABALLERO, AND E. M. R. A. ENGEL (2013): “Aggregate Implications of Lumpy Investment: New Evidence and a DSGE Model,” *American Economic Journal: Macroeconomics*, 5, 29–67.
- CARROLL, C. D. AND M. S. KIMBALL (1996): “On the Concavity of the Consumption Function,” *Econometrica*, 64, 981–992.
- DEN HAAN, W. J. (2010): “Assessing the accuracy of the aggregate law of motion in models with heterogeneous agents,” *Journal of Economic Dynamics and Control*, 34, 79–99.

- KAPLAN, G. AND G. L. VIOLANTE (2014): “A Model of the Consumption Response to Fiscal Stimulus Payments,” *Econometrica*, 82, 1199–1239.
- KHAN, A. AND J. K. THOMAS (2008): “Idiosyncratic Shocks and the Role of Non-convexities in Plant and Aggregate Investment Dynamics,” *Econometrica*, 76, 395–436.
- KRUSELL, P. AND A. A. SMITH, JR. (1998): “Income and Wealth Heterogeneity in the Macroeconomy,” *Journal of Political Economy*, 106, 867–896.
- LEE, H. (2025): “Global Nonlinear Solutions in Sequence Space and the Generalized Transition Function,” Working Paper.
- PETRONGOLO, B. AND C. A. PISSARIDES (2001): “Looking into the Black Box: A Survey of the Matching Function,” *Journal of Economic Literature*, 39, 390–431.
- SCHENK-HOPPÉ, K. R. AND B. SCHMALFUSS (2001): “Random fixed points in a stochastic Solow growth model,” *Journal of Mathematical Economics*, 36, 19–30.
- STOKEY, N. L., R. E. LUCAS, AND E. C. PRESCOTT (1989): *Recursive Methods in Economic Dynamics*, Harvard University Press.